An Identified-Model, Estimator-Consistency and Requirements Re-Baseline of a One-Axis CubeSat ADCS Testbed

A re-analysis of the SPACE 584 Fall 2025 final report

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Abstract

The SPACE 584 FA2025 class built a 1U-class CubeSat with a reaction wheel, two magnetorquers, a star tracker, a thermal-array sun sensor and a rotating test platform, and reported that most of its top-level goals were missed. This paper re-examines that work using recomputed numbers, a first-principles model identified from the report's own data (digitised from its raster figures), a redesigned estimator and controller evaluated by deterministic Monte Carlo, and a re-baselined requirement set. The audit found 29 items (4 critical, 15 high). Among them are an inertia error of a factor 2.6 that propagates into the wheel-resolution requirement, which is itself off by a factor of ten in units; a detumble requirement that bearing friction alone satisfies (28 s); a gyro that was saturated at its 500 dps range in the headline test; and a Helmholtz/torquer functional test that cannot distinguish magnetic action from friction. We identify the testbed friction (Coulomb torque 4.1×10^{-4} N m, validated to 10% on independent data) and show that the wheel channel was acceleration-limited rather than throttle-proportional. A gyro/sun/star EKF with a bias state and delay handling is NEES/NIS-consistent (mean NIS 1.004), whereas the report's filter is over-confident by about two orders of magnitude under realistic sensor noise. Replacing the state machine with a rate/attitude controller raises the simulated mission success within 60 s from 1% [0, 3] to 100% [98, 100], and a time-optimal slew is 13% [11, 14] faster than the braking-profile PID for 90°. A strict novelty screen against the literature reclassifies most candidate improvements as replications, and two of them (star-field design for uniqueness, coordinated wheel/torquer optimisation) did not hold and are dropped. The contribution that can be supported is a documented identified-model and estimator-consistency study of a student testbed, with a complete v2 requirement set, budgets and test campaign. Compared with current practice, the testbed's as-built accuracy (5.0° rms between its optical sensors) is that of the coarsest flight class, its bearing friction is 8–40× above balanced university tables, and its sun sensor is two orders of magnitude slower than a commercial unit. Sizing a flight version for a 400–500 km orbit with the SMAD methods shows that orbit disturbances are 147–7,169× smaller than the testbed friction, that 0.2 A m^2 torquers detumble the 2U from a 10°/s tip-off within about two orbits, and that the imaging payload, not the ADCS, drives power (about 10% duty) and data (one image every few days over UHF). All numbers regenerate from `run_all.py`. We then design a next-version bidirectional wheel, a stack of printed circuit boards as structural levels, and firmware, and test them in software-in-the-loop. In simulation the design stops within 0.030° (median; 95th percentile 0.064°) against 21.6° for the as-built wheel at its best, and the gain comes first from torque authority and from tracking the rotor angle through zero speed. The design has not been built, and the accuracy on the real testbed will be set by systematic errors that the simulation omits.

Nomenclature

Dynamics and actuators		*Estimation and sensing*	
θ, ω	body attitude [rad] and rate [rad/s]	b (EKF)	gyro bias state [rad/s]
I_z, I_{cs}	CubeSat inertia about the spin axis [kg m^2]	P, Q, R	state, process and measurement covariances
I_p, I_t	platform and total (loaded) inertia [kg m^2]	ν, S	innovation and its covariance
I_w, Ω	wheel inertia and speed	τ	measurement latency [s]
q	wheel momentum in body-rate units [rad/s]	NEES, NIS	normalised estimation error / innovation squared
T_c, b (friction)	Coulomb torque [N m], viscous coefficient	IFOV	instantaneous field of view [μrad]
τ_e, r_{max}	wheel lag [s], acceleration limit [rad/s^2]	t_e	exposure time [s]
m, B, μ_{eff}	dipole [A m^2], field [μT], effective permeability	s	smear [px]

1 Introduction

Attitude determination and control is one of the subsystems that most often limits a CubeSat mission, and a ground testbed is the usual way for a student team to exercise it before flight. In the Fall 2025 SPACE 584 course the class designed, built and tested a one-axis ADCS housed in a 1U-class structure, together with the star field, Helmholtz coil and rotating platform needed to test it. The final report [1] documents that work honestly, and also reports that most of the top-level goals were missed. The author contributed the CAD, the sensor testing and the state controller to that team report.

The report leaves open *why* the goals were missed. Several of its numbers can be recomputed from its own equations, and its figures contain the only measurements of the testbed's dynamics, but neither had been checked. The inputs to this study are the 53-page final report of SPACE 584 (FA2025) [1], its public code repository (branch `integrated_adcs`) [2], and the author's round-1 notes. The repository was cloned into an isolated folder and read as data only; no code from it was executed. Where the report's figures contain the only available measurements (Figs. 37 and 38), they were digitised from the raster images embedded in the PDF using tick-detected axis calibration (resolution $\approx$0.05 rad/s, 0.03 s).

The goal of this paper is to determine which of the report's conclusions survive a full recomputation, and to give a baseline that a v2 team could build and test. To do this, the study (1) audits the report (§3); (2) identifies a dynamic model of the platform, CubeSat and wheel from the report's data (§4); (3) redesigns the controller and estimator and tests them by Monte Carlo (§5–6); (4) re-examines star identification on the *real* star-field map (§7); (5) redesigns the magnetic subsystem and the test coil (§8); (6) analyses imaging and computes the spin-rate thresholds the report left open (§9); (7) screens candidate novelties against the literature (§10); (8) gives a v2 requirement set, budgets and test campaign (§11); (9) compares the testbed with commercial units, flown missions and university facilities (§12); and (10) sizes a flight version for a real LEO orbit and gives recommendations (§13–14).

The claims of this paper are stated strictly, as follows:

- *Replicated*: time-optimal slews, gyro-bias estimation from optical fixes, delayed-measurement updates, wheel/magnetorquer momentum management, star-ID voting and gyro-aided stacking. All of these exist in the literature (§10); here they are quantified for this testbed.
- *Improved over the report*: the arithmetic and unit corrections, the identified model, a consistent estimator, the controller, the thresholds, the requirement set, the drive electronics and the coil geometry.
- *Novel (supportable only)*: a documented *identified-model plus estimator-consistency study of a student testbed*, built from the testbed's own digitised data, with its provenance and uncertainty. We claim no new algorithm.
- *Dropped*: star-field design for uniqueness, the coordinated LQ wheel/torquer scheme as an improvement, and the faster-star-ID claim.

The status of every input is tracked as VERIFIED (read from a datasheet or paper), DERIVED (computed from verified values), or UNVERIFIED/assumed (flagged in the text and in `research/params.json`).

The rest of the paper is structured as follows. Section 2 describes the approach to the data, its provenance and the lumped CAD model. Section 3 presents the audit of the report, and Sections 4–9 develop the identified model, the controller, the estimator, star identification, the magnetic subsystem and imaging. Section 10 screens the candidate improvements against the literature, and Section 11 gives the v2 baseline and test campaign. Section 12 compares the testbed with current practice, Section 13 takes the design from the

testbed to an orbit (environment, disturbances, actuator sizing, detumble, power, data, link and verification), and Section 14 gives recommendations for a flight design. The paper then discusses the lessons learned and the sources of error and model limitations (Section 16), and concludes with an assessment of what the testbed can and cannot demonstrate (Section 17).

2 Approach: Data, Provenance and the Lumped CAD Model

Every number in this paper is either read from a source, derived from read values, or assumed, and the analysis is built so that each can be regenerated from the report and the data. Two inputs underpin most of what follows: the digitised measurements and a lumped CAD model for the inertia.

Digitisation. Fig. 38 (rate, throttle, attitude) and Fig. 37 (ST/SS angle) of the report are embedded as rasters (2264×1590 and 2200×535 px). Axis ticks are detected from the frame, a linear map is fitted (residual <2 px), and the blue and orange series are tracked by colour. The gyro curve resolves ≈0.05 rad/s. The Hall curve is piecewise linear between sparse pulses (one per revolution), so it is used only as an independent check.

Lumped CAD model. The layer stack (PETG base 4 mm, three 1.6 mm Al plates, four M2 brass standoffs, 100×150 mm) is built in `cadquery`. The solid inertia comes from the CAD solids, and the component masses are catalogue values (flagged as assumed), with the battery mass taken as the remainder to 960 g. That remainder is ≈235 g, i.e. one 4S pack (Fig. 1). A Monte Carlo over ±15% in component mass and position gives the I_z band quoted in §3 and a CoM offset of 2.9 mm.

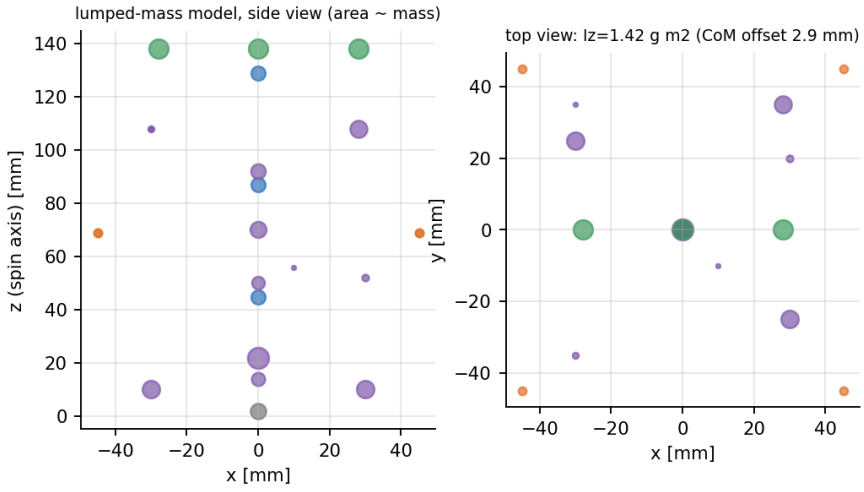


Figure 1: Lumped-mass model used for the inertia estimate (marker area ∝ mass). The battery is the remainder to the measured 960 g.

3 Audit of the Report

The audit is generated by `audit/audit_checks.py` and lists, for each item, the claim, the paper value, the recomputed value, the effect and a fix. Table 1 gives the ranking, and the most consequential findings are developed below. Figure 2 summarises the count by severity together with the v2 budgets used later.

3.1 Headline Findings

A01 Inertia (critical). The report sizes the wheel with $I_{sat} = 0.5 \cdot 1.5 \cdot 0.05^2 = 0.00375$ kg m^2. The product is in fact 0.001875, a factor-2 arithmetic slip. The model also uses 1.5 kg for a 0.96 kg spacecraft, and a disk for what is a square prism. A parametric CAD/lumped-mass model (`cad_inertia.py`, 960 g total, battery mass as the remainder) gives $I_z = 1.42 \times 10^{-3}$ kg m^2 (5–95%: 1.38–1.46), against closed-form bounds of 1.2–1.6 (disk, 10×10 prism). The wheel therefore needs 133 rad/s (1270 RPM), not 312 rad/s, to absorb 1 rev/s, and the inertia ratio is ≈19–21, not 49.6.

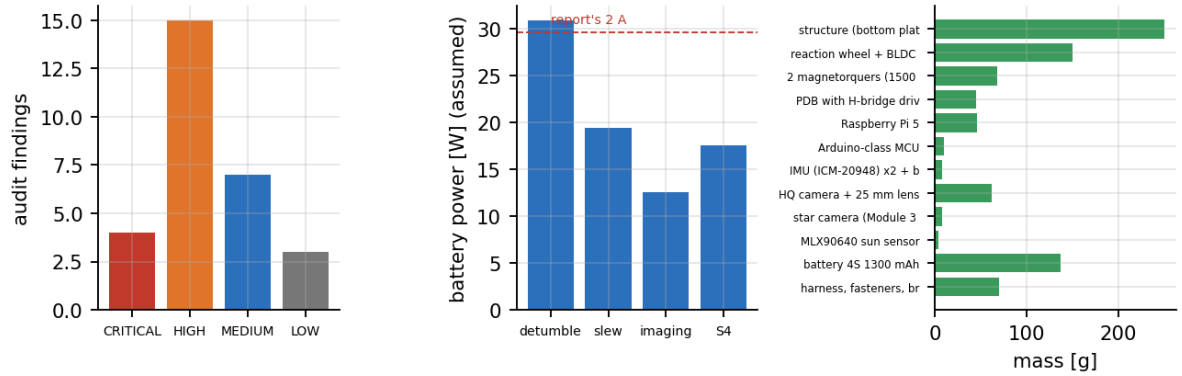


Figure 2: Audit findings and v2 budgets. Left: audit findings by severity. Middle: v2 battery power per mode (assumed loads) against the 2 A the report sized for. Right: v2 mass budget.

A02 Resolution requirement (critical). A rate of 0.001 rad/s is 0.00955 RPM, not 0.0942 RPM, so the requirement is ten times too large. With the corrected ratio the wheel step must be ≈ 0.2 RPM. The 10-bit ESC step of 13.67 RPM is $\approx 70\times$ too coarse, and the proposed 12-bit step (3.42 RPM) is still $\approx 17\times$ too coarse, which contradicts the report's conclusion. About 16 bits (or a closed speed loop) removes the issue. There is, however, a nuance found in §5: a fixed-step state machine cannot settle below one bin, but closed-loop rate feedback dithers across bins and still reaches a few mrad/s rms on the same coarse wheel. Resolution is therefore as much a control-logic problem as a hardware one.

A03 Requirement A is satisfied by friction (critical). The platform stops a mounted CubeSat from 1 rev/s in 37.8 s with no ADCS (Table 12 of the report), and the identified friction brings it below 10°/s in 28 s. As a result, the requirement "1 rev/s→10°/s in 60 s" cannot discriminate a working ADCS from a dead one.

A04 Gyro saturation (critical). In Fig. 38 the gyro trace is flat at -8.70 rad/s for $t < 11$ s, while the Hall sensor reaches -10.6 rad/s. Since 500 dps equals 8.727 rad/s, and the team's sketch sets `ICM20948_GYRO_RANGE_500_DPS` [2, 3], the gyro was saturated. The test started at 1.7 rev/s, not 1 rev/s, and the first 10 s of rate feedback was clipped (Fig. 3). The text's "up to 2.1 rad/s in the opposite direction" also disagrees with the plotted peaks of 2.81 (gyro) and 2.69 (Hall).



Figure 3: Spin rate versus time in the report's headline test (Fig. 38, digitised), with the gyro's 500 dps full scale. The gyro is clipped for the first ≈11 s, so the rate feedback was wrong when the controller needed it most. Data digitised from the report [1]; model from Section 4.

A07/A08 Smear budget. The rate requirement uses 0.5×0.0002 rad, but 0.0002 rad is the 1 mm resolution (3.2 px). The HQ-camera IFOV with the 25 mm lens is 62 μrad (1.55 μm pixels [4]), so 0.5 px is 31 μrad, i.e. 3.1×10^{-4} rad/s at the same 0.1 s exposure and 1.9×10^{-3} rad/s at 1/60 s. The test images taken at 0.5–2.5 rad/s carried 134–672 px of smear at 60 fps, about 269× the budget. The failure of the 1 mm objective is therefore explained by motion, and the statement that the 'focus shifted when in motion' is not

supported.

A09–A11 Magnetorquers. A relay is an on/off device with no bipolar drive or flyback path. The 493-turn rod is 3.4 Ω, draws 1.47 A at 5 V and stores 6.0 mH. The report's thermal threshold (0.01 K/s at 9 V) is off by a factor of ∼200: the recomputed heating is 0.70 K/s at 5 V and 2.3 K/s at 9 V. Because $m = \mu_{eff} NIA$ with $I = V/R \propto 1/N$, the dipole is independent of the number of turns at fixed voltage. The rewinding to ≈3400 turns on a shorter rod therefore lowered the modelled dipole from 1.14 to 0.68 A m^2 and raised the mass from 22 to 61 g.

A13 De-saturation requirement (high). A wheel speed of 200 rev/s (12 000 RPM) exceeds the wheel's own 7000 RPM, and removing the stated 0.090 N ms in 120 s needs 0.75 mN m, i.e. 8.4 A m^2 at 90 μT, against the 0.95–1.6 delivered. The target is physically infeasible for this hardware.

A14–A16 Helmholtz coil. A square pair needs a separation of 0.546 of the side. The built 14 in separation on 12 in coils gives a 26% peak-to-peak variation over the body (10% at the right spacing). The stated wire gives 83 Ω and 0.06 A at 5 V, but 0.97 A was measured. Table 10's field falls by 63% (factor 2.7) between sensors 39 mm apart, the resistance rises by 19% (≈39 K), and the cross-axis coupling is 15%. The functional test ("stopped in 50 s") is not evidence of magnetic detumbling: friction stops the loaded platform in 38 s, and the available magnetic torque is only 0.21 of the friction torque.

A18 Power. A current of 2 A for 3 min is 100 mAh, so "over 2200 mAh" is 22× what the arithmetic requires. Three 4S 2200 mAh packs weigh 717 g (verified pack masses 232–245 g [5]), and the lumped mass budget leaves room for only one. The 30 s run time is therefore not a capacity problem; a brown-out is the likely, but undiagnosed, cause.

A20–A22 Sensors and estimation. The digitised Fig. 37 shows one star-tracker and one sun-sensor sample every 6.4 s (0.16 Hz, not the stated 0.5 Hz), 15 distinct sun-sensor levels in 160 s, and an RMS ST–SS discrepancy of 5.0° (max 9.5°). The EKF's 5 nT magnetometer noise corresponds to 0.010°, whereas one LSB of the ICM-20948 magnetometer (0.15 μT [3]) is 0.30°. Details are given in §6 and §7.

Table 1: Audit findings ranked by severity (full text, paper vs recomputed values and fixes in `audit/audit.md`).

ID	Severity	Area	Finding (short)
A01	Critical	Reaction wheel sizing	The product 0.5*1.5*0.05^2 is 0.001875, not 0.00375 (a factor-2 arithmetic slip).
A02	Critical	Reaction wheel resolution	Unit conversion error: 0.001 rad/s is 0.00955 RPM, ten times smaller than the 0.0942 RPM used.
A03	Critical	Detumble requirements / test validity	The testbed's own bearing friction (identified from Tables 11-12, torque ~4e-4 N m) brings a mounted CubeSat from 1 rev/s to 10 deg/s in about 28 s and to rest in 38 s with the ADCS off.
A04	Critical	Gyroscope / attitude control test	In Fig. 38 the gyro trace is flat at -8.70 rad/s for t<11 s while the Hall sensor continues to rise to -10.6 rad/s.
A05	High	Attitude control test	The plotted peak is 2.8 rad/s (gyro) and 2.7 rad/s (Hall) at t = 45 s, 30-35 % above the quoted value..
A06	High	Attitude control test	Momentum bookkeeping on the gyro trace (friction removed) shows the wheel accelerating at a nearly constant ~0.3 rad/s^2 (body-equivalent) regardless of the throttle step (sensitivity 0.04 rad/s^2 per 100 % throttle, bootstrap 95 .
A07	High	Camera / smear budget	0.0002 rad is the 1 mm resolution requirement (3.2 px), not the pixel IFOV.
A08	High	Camera / imaging test	At 0.5 rad/s and a 60 fps exposure the smear is 134 px (about 40 mm on the target) against a 0.5 px budget: 270x over.
A09	High	Magnetorquer drive electronics	A relay is an on/off device: it cannot realise a proportional dipole from 'varying PWM', has no bipolar drive (B-dot needs both polarities), contact bounce/wear at PWM rates, and without a flyback path the ~6 mH coil (0.01 J scale.
A10	High	Magnetorquer thermal sizing	With R = 3.4 Ω (0.34 Ω/m x ~10 m), P = V^2/R and a heat capacity of ~11 J/K (rod 17.8 g steel + wire 5.6 g Cu), heating is 0.7 K/s at 5 V and 2.3 K/s at 9 V, 70-230x the quoted values.

ID	Severity	Area	Finding (short)
A11	High	Magnetorquer sizing	At fixed voltage the dipole m = mu_eff*N*I*A is independent of N (I = V/R and R grows with N), exactly as the report itself shows for the Helmholtz coil.
A13	High	Momentum dump requirement	The wheel as built cannot reach 12000 RPM (the report's own limit is 7000 RPM), and dumping the stated momentum in 120 s needs 0.75 mN m, i.e. 8.4 A m^2 at the 90 uT test field: 5-9x more than the torquers deliver.
A14	High	Helmholtz coil design	Square Helmholtz pairs need a separation of 0.5445-0.546 of the side (here 6.6 in); 14 in is 2.1x too wide and gives ~26 % peak-to-peak variation over the 10 x 10 x 15 cm body (10 % at the right spacing).
A15	High	Helmholtz calibration (Table 10)	The field falls by 2.7x (63 %) between sensors 39 mm apart, as for a near-wire field (~1/r^1.8), not as in the uniform centre of a 305 mm coil; so the sensors were not at the centre, or the coil is not uniform.
A16	High	Helmholtz functional test	Bearing friction alone stops the loaded platform from 1 rev/s in 38 s (Table 12).
A18	High	Power / battery sizing	2 A for 3 min is 100 mAh; the 2200 mAh figure is 22x the report's own arithmetic.
A20	High	Sun sensor / star tracker accuracy claims	Digitised markers in Fig. 37 show one SS sample every 6.4 s and one ST sample every 6.4 s (0.16 Hz), not 0.5 Hz.
A21	High	EKF assumptions and validation	5 nT of magnetometer noise corresponds to 0.010 deg in heading but the ICM-20948 magnetometer resolution is 0.15 uT (0.30 deg): the assumed noise is 30x finer than one LSB.
A22	High	Star identification / artificial field	On the real 56-star map, 99 % of pair distances have a coincident partner within the 0.004 rad tolerance and 92 % of stars share their first-neighbour distance with another star, so the unique-match claim rests entirely on later c.
A12	Medium	Magnetorquer test	0.95/1.6 = 59 % of the target, i.e. 41 % low.
A17	Medium	Rotating platform data (Tables 11-12)	Spin-down times are not monotone in speed in Table 11 (0.84 -> 4.9 s, 1.04 -> 7 s, 1.39 -> 7.3 s) and the best-fit Coulomb model leaves ~20 % log residuals.
A19	Medium	Camera resolution	The static claim is consistent: 0.7 mm = 2.3 px of the 0.31 mm GSD (Nyquist limit 0.62 mm), FOV 14.3 x 10.8 deg agrees.
A23	Medium	Requirements table	(i) 200 rev/s (12000 RPM) exceeds the 7000 RPM / 117 rev/s wheel by 1.7x; (ii) Table 1 'slow to 30 deg/s' and Table 6 'less than 0.5 deg/s after 2 minutes using wheels and rods' are mutually inconsistent in time bases (60 s vs 2 m.
A24	Medium	State machine / thresholds	The thresholds follow from exposure, IFOV and update period (a6): sun-sensor acquisition <= 0.38 rad/s at 2 Hz updates but only 0.04 rad/s at the measured 6.6 s period; star tracker <= 3.9 mrad/s (1 px, 0.1 s) to 0.058 rad/s (3 px.
A25	Medium	Mass / volume	150 mm is 1.3-1.5U, not 2U (2U = 227 mm); the 1 kg limit is a course constraint, not the CubeSat spec (Rev 14.1: 2 kg for 1U); the two mass figures differ by 10 g without a stated measurement uncertainty; the volume requirement wa.
A26	Medium	Hall sensor / Fig. 38 agreement	With one Hall pulse per revolution the sensor gives ~1 sample per 2.5-12 s once the platform slows; the orange curve is a straight-line interpolation (vertices visible) and differs from the gyro by up to 1 rad/s between vertices.
A27	Low	State numbering / telemetry	The two enumerations disagree (S3 is science imaging in one and desaturation in the other); Table 13 lists 33 doubles (indices 0-32) but states 35 doubles/280 bytes..
A28	Low	Cost / schedule	$6000 - $3000 leaves $3000, not a $4000 margin; the requirement as written is ambiguous..

ID	Severity	Area	Finding (short)
A29	Low	Wheel speed arithmetic	311.6 rad/s is 2976 RPM; 'KeV' should be KV.

4 Identified Platform and Wheel Model

4.1 Bearing Friction

For a Coulomb friction torque T_c and a viscous coefficient b, the time to stop from ω_0 is

$$t_s = \frac{I}{b} \ln\left(1 + \frac{b\,\omega_0}{T_c}\right), \tag{1}$$

or $I\omega_0/T_c$ for $b = 0$, where I is the rotating inertia. The report's Tables 11–12 give nine stop times (six unweighted, three with the CubeSat). We fit T_c, b and the platform inertia I_p (with I_{cs} from CAD and its $\pm$ range of 1.2–1.6 g m^2 propagated) by least squares on the log-times. The Coulomb-only model is preferred by AICc, with $T_c = 4.1 \times 10^{-4}$ N m (95% bootstrap 2.7–6.9), $I_p = 0.45 \times 10^{-3}$ kg m^2 and a loaded inertia of 1.87×10^{-3}. The residual is 24% in log-time: the data are noisy (non-monotone in Table 11) and a viscous term cannot be separated. As an independent validation, the model applied to the pre-wheel segment of Fig. 38 (Hall rate $-10 \to -7.8$ rad/s) predicts a deceleration of 0.221 against 0.200 rad/s^2 measured (Fig. 4).

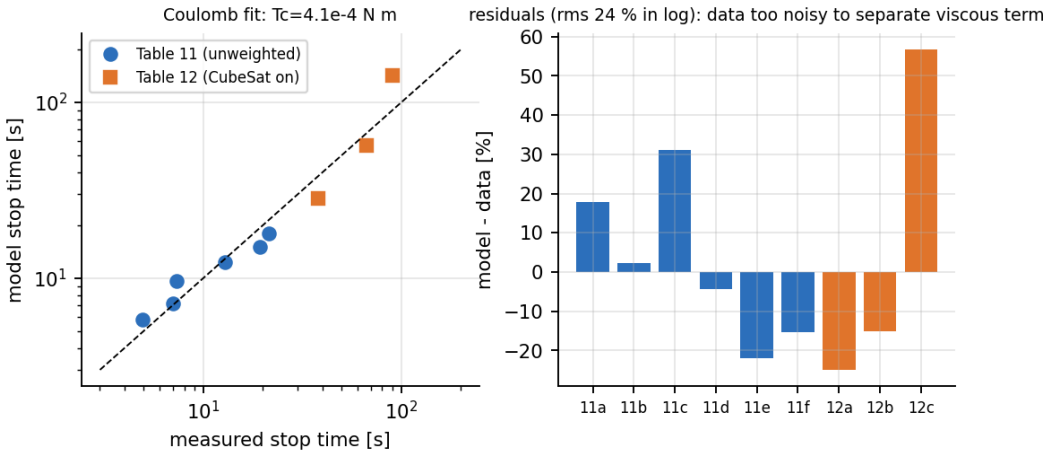


Figure 4: Platform spin-down model versus the report's stop-time data. Left: model versus measured stop times. Right: relative residuals for Table 11 (blue) and Table 12 (orange).

4.2 Wheel and ESC Channel

Let $q = -I_w\Omega/I_t$ be the wheel momentum in body-rate-equivalent units. Momentum conservation then gives

$$\dot{\omega} = \dot{q} - f\,\mathrm{sgn}(\omega), \tag{2}$$

where $f = T_c/I_t$. The *model-free* momentum trace $q(t) = \omega(t) - \omega(t_0) + \int f\,\mathrm{sgn}\,\omega\,dt$ shows (Fig. 5) that the wheel gains momentum at an almost constant 0.31 rad/s^2 over the whole staircase, and still holds 8.4 rad/s-equivalent at 60 s, although the report states it was 'off'. A parametric fit with a throttle-sensitivity term s gives $s = 0.04$ rad/s^2 per unit throttle (95% block-bootstrap [-0.05, 0.14]; ΔAICc=5.4), which is consistent with throttle having no effect, with a residual rms of 0.25 rad/s. The first '0%' step had already consumed most of the momentum authority ($\tau = 0.42$ s transient). The equivalent wheel torque is 5.8×10^{-4} N m and the peak held momentum is 13.2 rad/s-equivalent (≈ 3115 RPM if I_w and I_t are right). This directly shows that the wheel was acceleration-limited and that the throttle-to-speed map was never exercised.

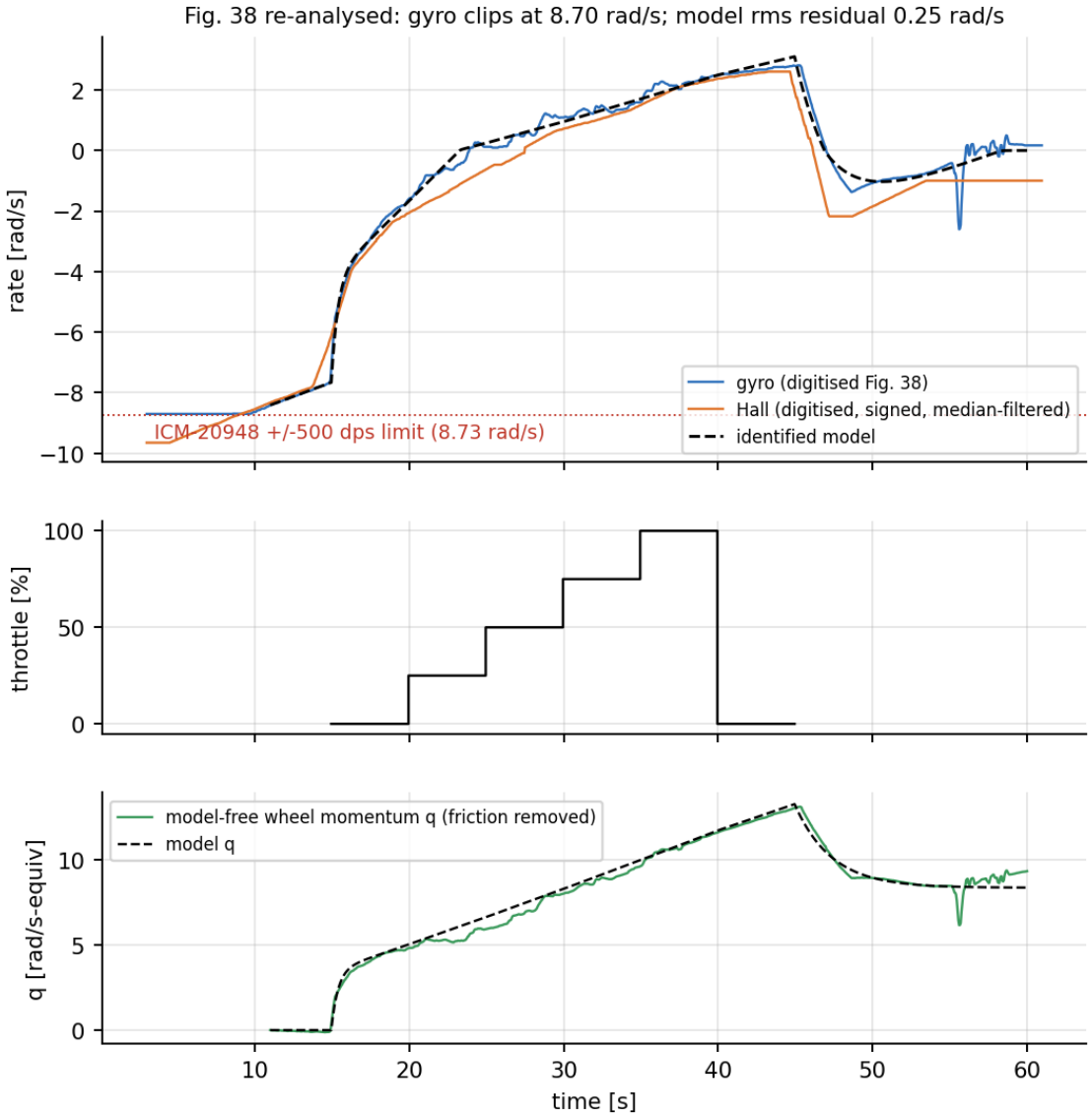


Figure 5: Re-analysis of the report's Fig. 38. Top: digitised gyro (clipped at −500 dps), Hall and identified model. Middle: throttle staircase. Bottom: wheel momentum, model-free and model. The momentum rises at a near-constant rate regardless of throttle.

5 Controller Redesign

5.1 Plant, Sensors and Estimator

The simulator (`simlib.py`) integrates

$$\dot{\theta} = \omega, \qquad \dot{\omega} = \dot{q} - f\tanh(\omega/0.03) + d, \tag{3}$$

where d is a disturbance torque per unit inertia, with the wheel modelled as $\dot{q} = \mathrm{clip}((q_{cmd} - q)/\tau_e, \pm r_{max})$, $|q| \le q_{max}$, and a command resolution Δq. For the v1 hardware we use $\Delta q = 0.058$ rad/s-eq (13.67 RPM), $r_{max} = 0.31$ rad/s^2 (identified), a gyro range of ±500 dps, and optics every 6.6 s with a 4 s star-tracker delay and a 6% misidentification rate. For v2 we use $\Delta q = 4 \times 10^{-4}$, $r_{max} = 0.53$ (1 mN m), a gyro range of ±1000 dps (a verified option [3]), and 2 Hz optics with 0.3 s latency. The gyro noise is 0.015 dps/$\sqrt{\text{Hz}}$ over 51 Hz [3], the initial zero-rate offset is $\sigma = 3$ dps (datasheet ±5 dps), and a bias drift of 5×10^{-5} rad/s/$\sqrt{\text{s}}$ is assumed. Optical fixes are gated by the fields of view (sun pad at −12°, ±27.5°; star field −66° to 8° plus the crop) and by blur (the star tracker is unusable above 0.02 rad/s), and their noise includes smear. All controllers share the same EKF (§6) and a search mode (a sweep at 0.3 rad/s when the attitude is unknown), which the report lacked.

5.2 Controllers

Four controllers are compared.

Report state machine (ported): S0 is gyro-only rate stepping by one wheel bin per 0.5 s tick, S1 is coarse pointing at < 0.3 rad/s, S2 is fine pointing and S3 is imaging, with transition thresholds of 0.3 and 0.05 rad/s as in the report's Fig. 8.

Cascade PID: the attitude loop commands $\omega_{ref} = -\mathrm{sgn}(e)\min(K_\theta|e|, \sqrt{2 \cdot 0.6\, a_{max}|e|})$ (a braking-capable profile), followed by a rate PI with back-calculation.

LQR/LQG: $u = -K[\theta_e, \omega, \int \theta_e]$, obtained from the algebraic Riccati equation for the triple integrator with a torque-limit weight $R = 1/a_{max}^2$ and the error clipped for saturation.

Time-optimal: bang-bang control on the switching curve $s = e + \omega|\omega|/(2a) + \omega(\tau_e + 0.04)$ with $a = 0.9\, a_{max}$, handing the hold phase to the LQR.

The gains were tuned on a 30° slew and hold, using a grid over $K_\omega \in \{2, 5, 10\}$, $K_\theta \in \{0.5, 1, 2\}$ for the PID and over weight/effort ratios for the LQR. The selected values are $K_\omega = 5$ and $K_\theta = 2.0$, and the LQR gain is $K = [3.47, 3.08, 0.32]$.

Stability margins. For the loop with a wheel lag of 0.05 s and a 50 ms delay, the PID rate loop has a phase margin of 52° and a gain margin of 13 dB, the attitude loop 63°/11 dB, and the LQR 52°/49 dB. Anti-windup is by back-calculation (PID) and by clipping of the error and integrator (LQR), and the momentum limit is enforced in the wheel model.

5.3 Results

Table 2 gives the full-mission Monte Carlo ($N = 300$ initial conditions per cell, seed-fixed). With the as-built wheel, the report's state machine succeeds in 1% [0, 3] of trials (rms rate 13.6 mrad/s against the 6.2 mrad/s imaging limit). One wheel bin (0.058 rad/s) exceeds the S1 cut-off and the imaging limit is 9× below one step, so the logic cannot converge. A properly retuned PID on the *same* hardware reaches 26% [22, 32], and v2 hardware reaches 100% [98, 100] (PID), 97% [95, 99] (LQR) and 99% [97, 99] (time-optimal), with median ready times of 21.4 [20.4, 22.5], 24.7 [22.9, 26.4] and 21.2 [19.7, 22.3] s. With the attitude known (as in the report's own Simulink test) and friction removed, the report's logic with 13.7 RPM bins reaches imaging in 1% [0, 4] of trials, a v1-resolution PID in 95% [90, 97] with an rms rate of 3.6 mrad/s, and v2 hardware in 100% [98, 100] (rms 4.6 mrad/s). In the v1 case the feedback dithers across the wheel bins, so quantisation alone is not fatal to a closed-loop controller; it is fatal to a fixed-step state machine. Figures 6 and 7 show example trajectories and the success statistics.

Table 2: Full-mission Monte Carlo (N=300, random initial spin 0.5–1.7 rev/s and angle, EKF + delayed optics + search mode, test-bed friction on). 'Ready' = within ±3.5° and below 6.2 mrad/s for 1 s. Median [95% bootstrap CI]; success is Wilson 95%.

Hardware	Controller	P(ready≤60 s)	t_{ready} median [s]	rms ω last 10 s [mrad/s]
v1	report SM	1% [0, 3]	22.8 [14.3, 27.0]	13.6
v1	PID (v1 gains)	26% [22, 32]	62.9 [60.6, 64.6]	28.4
v2	cascade PID	100% [98, 100]	21.4 [20.4, 22.5]	0.7
v2	LQR/LQG	97% [95, 99]	24.7 [22.9, 26.4]	0.6
v2	time-optimal	99% [97, 99]	21.2 [19.7, 22.3]	1.5

Table 3: Rest-to-rest slew settle time (air-bearing plant, attitude known to 0.3°, N=200). Median [95% CI] and overshoot; $t_{min} = 2\sqrt{\theta/a}$ is the bang-bang bound for $a = 0.9\, a_{max}$.

Slew	PID [s]	LQR [s]	time-optimal [s]	TO gain vs PID	t_{min} [s]
30°	4.18 [4.12, 4.21]	3.06 [3.06, 3.08] (ov 0.5°)	3.58 [3.54, 3.60]	14% [13, 15]	2.1
90°	5.90 [5.80, 6.00]	5.74 [5.73, 5.76] (ov 3.5°)	5.13 [5.10, 5.18]	13% [11, 14]	3.6
150°	6.86 [6.78, 6.95]	7.04 [7.03, 7.04] (ov 5.2°)	6.18 [6.16, 6.22]	10% [9, 11]	4.7

Detumbling from 1–1.7 rev/s is torque-limited and nearly identical for every controller (≈15 s). The differences appear in rest-to-rest slews (Table 3): the time-optimal controller is 13% [11, 14] faster than the PID at 90° and 10% [9, 11] faster at 150°, and 11% [10, 11] / 12% [12, 12] faster than the saturated LQR, which

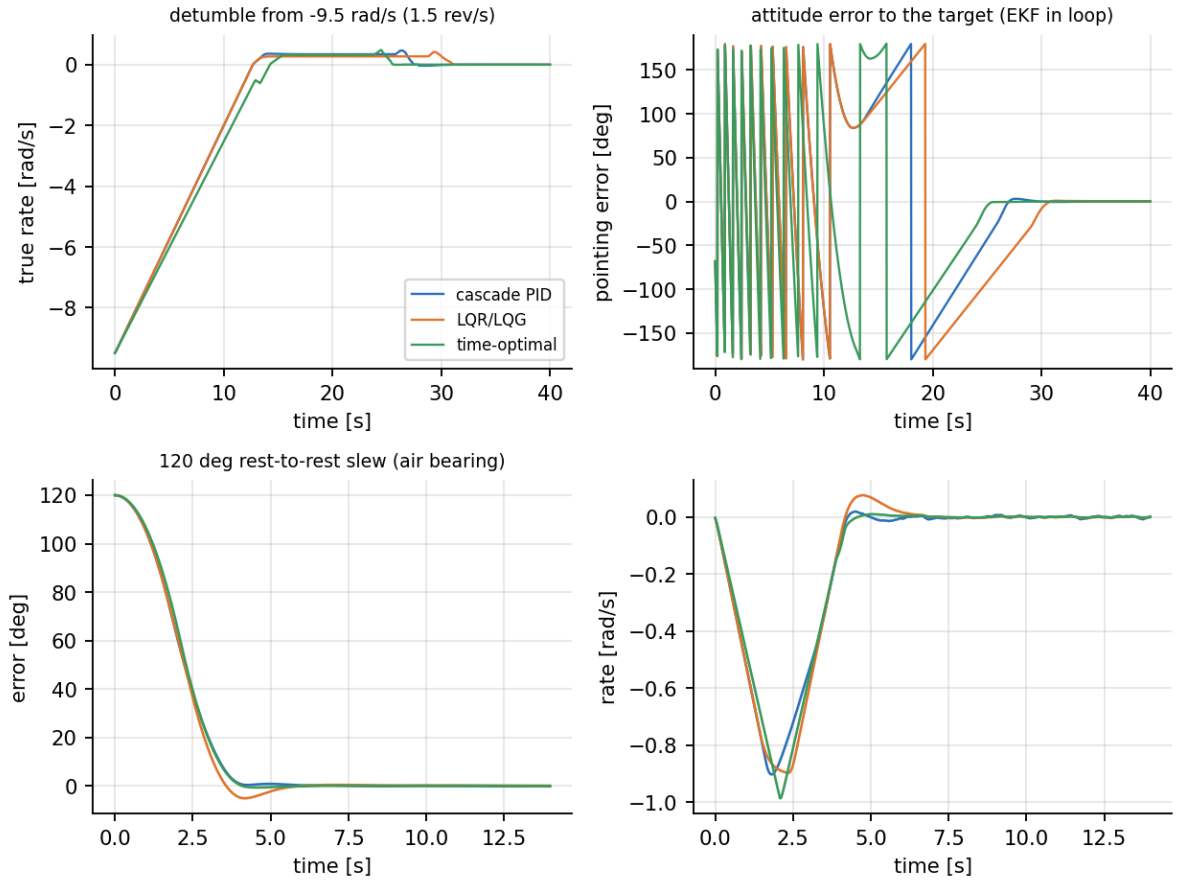


Figure 6: Example trajectories with the same initial conditions and the EKF in the loop. Top: detumble from −1.5 rev/s. Bottom: 120° rest-to-rest slew on an air-bearing plant.

overshoots by 3.5° at 90°. Against a 3-s, 0.4 rad/s^2 step disturbance on an air bearing, the peak deviation is 1.7° (PID), 6.2° (LQR) and 3.0° (time-optimal).

Delay and update-rate sensitivity. We sweep the optics period (0.15–6.6 s) and latency (0.3 or 3 s) for the LQR and time-optimal controllers (N=150 each; `results/a2_control.json`, 'optics_sensitivity'). With 0.3 s latency, periods up to 2 s still give ≥95% success within 60 s, but the measured 6.6 s period gives only about 40%, because the gyro cannot carry the attitude between sparse, intermittently visible fixes. A 3 s latency reduces success to roughly 30–95%, depending on the period. The latency rows are *not* monotonic in the update rate (fast updates do worse than 2 s updates at 3 s latency), because the correction of fixes still in flight is approximate. They should therefore be read as a warning about long latency, not as design data. It should also be noted that the report's 0.5 Hz figure is not what was measured (6.4 s between fixes).

5.4 The Thresholds the Report Left Undetermined

Table 4 computes these thresholds, and Figure 8 compares them. For imaging, the limit is $\omega \leq 0.5\,\mathrm{IFOV}/t_e$, which gives 6.2 mrad/s at 5 ms (1.9 mrad/s at 1/60 s). The star tracker needs a blur of $\lesssim 3$ px of 3.9×10^{-4} rad, i.e. 0.058 rad/s at 20 ms (3.9 mrad/s at 1 px, 100 ms). The sun sensor must keep the pad inside its 55° field between updates, which allows 0.38 rad/s at 2 Hz but only 0.043 rad/s at the measured 6.6 s update period. The report's 0.001 rad/s (A07) is therefore not a consistent bound.

6 Estimation

6.1 The Report's Filter Under a Consistency Test

The report's two-state EKF $[\theta, \omega]$ uses a gyro plus $B\cos\theta$ and $B\cos(\theta+\pi/2)$ magnetometers every 2 s, with noise values described as 'arbitrary' (gyro 1 dps, 5 and 6 nT); it was re-implemented here. The normalised

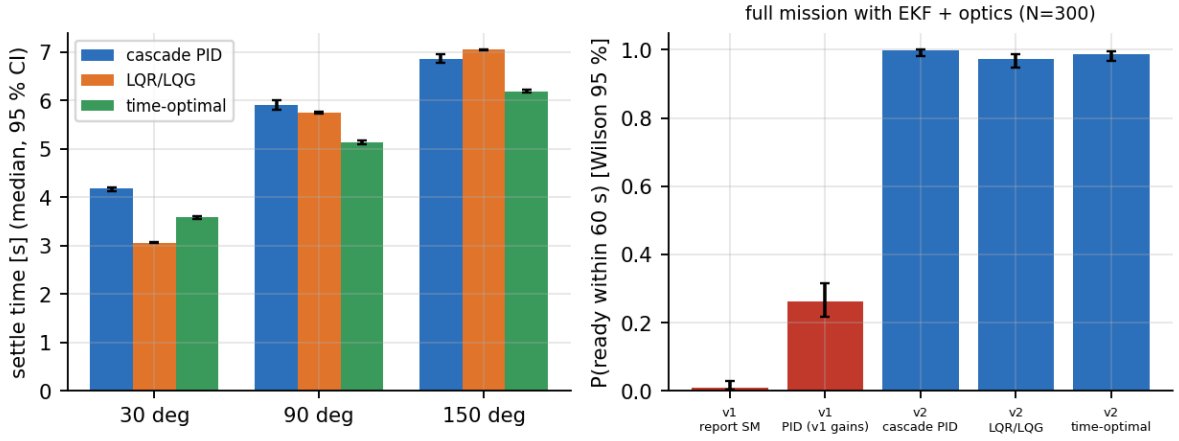


Figure 7: Slew and mission statistics for the controllers. Left: slew settle times (median, 95% CI). Right: probability of being imaging-ready within 60 s for the five hardware/controller combinations of Table 2.

Table 4: Spin-rate thresholds the report left undetermined (computed from IFOV, pixel scale, exposure and update period).

Quantity	Limit [rad/s]	Basis
S0→S1: sun-sensor acquisition	0.3000	sweep rate with smear-aware sigma; FOV limit 0.38 at 2 Hz updates, 0.04 at the measured 6.6 s
S1→S2: star tracker usable	0.0580	3 px blur at 20 ms exposure (0.0039 at 100 ms, 1 px)
S2→S3: imaging	0.0062	0.5 px smear at 5 ms (0.0019 at 1/60 s)
report derivation	0.0010	0.0001 rad / 0.1 s (inconsistent, A07)

estimation error squared

$$\epsilon_k = (\hat{x} - x)^\top P^{-1} (\hat{x} - x), \tag{4}$$

averaged over N independent runs, must lie in $[\chi^2_{Nn}(0.025), \chi^2_{Nn}(0.975)]/N$ [16], where n is the state dimension. For $n = 2$ and $N = 200$ the interval is $[1.83, 2.18]$. Even when the truth equals the report's own assumptions, the mean NEES is 0.32 (pessimistic by $\approx 6\times$). With ICM-20948 gyro noise and a 2-LSB (0.3 μT) magnetometer it is 187 (grossly over-confident), and 188 with a 3 dps bias. Its 5 nT magnetometer noise implies 0.010° of heading noise, which is 0.30° below one sensor LSB. The statement that "errors stay within the sigma bounds" is not a consistency test.

6.2 Redesigned Filter

The redesigned filter uses the state $x = [\theta, b]^\top$ and is driven by the gyro:

$$\hat{\theta}^+ = \hat{\theta} + (g - \hat{b})\Delta t, \qquad F = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix}, \qquad Q = \mathrm{diag}(n_d^2 \Delta t, \sigma_b^2 \Delta t), \tag{5}$$

where g is the gyro reading and n_d is the gyro angle-random-walk density. Sun-sensor and star-tracker fixes z (period 0.5 s, latency τ) are applied by the *measurement-advance* method, with innovation $\nu = z - \hat{\theta}(t - \tau)$ and $S = P_{11} + R + n_d^2 \tau + \tau^2 P_{22}$. The delay is thus handled with the stored estimate at the measurement epoch, and the extrapolation uncertainty enters R (conservative, and positive-definite by construction; cf. [14, 15]). Fixes are gated at 6σ, the filter is re-initialised after four consecutive rejections, and R includes the motion smear $\sqrt{(\omega t_{int})^2/12}$.

Two implementation errors found by the consistency tests are worth recording. (a) A first version that applied an extrapolated-gain covariance decrement lost positive-definiteness, and was replaced by the form above. (b) Fixes still in flight were compared with stored epochs that did not yet contain the corrections already applied, which double-counted information; before the fix, a 3 s latency gave 6% mission success and a mean NEES of order 10^4. The corrections are now carried into the stored epochs, and on (re-)initialisation the stored epochs are re-based on the new fix using the integrated gyro.

The filter was evaluated by Monte Carlo ($N = 500$, seed-fixed, with a 0.3 rad/s sweep so that fixes arrive intermittently). The mean NEES is 2.13 against [1.83, 2.18] (77% of time steps inside the interval), and the

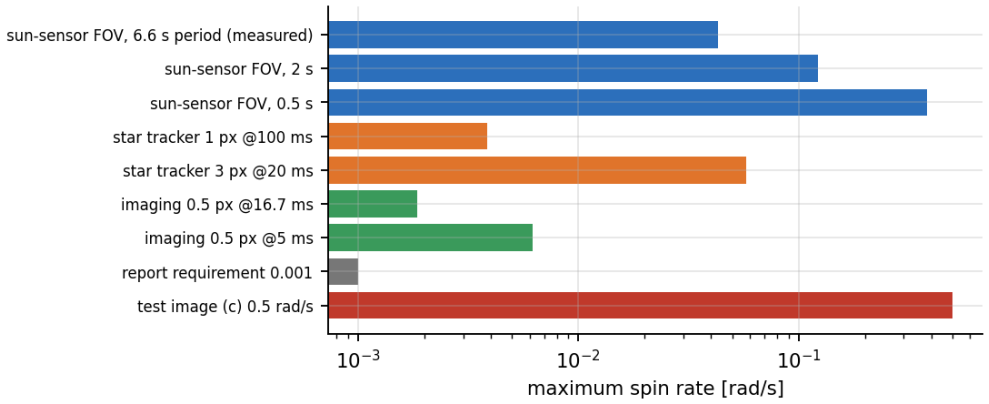


Figure 8: Maximum spin rates implied by each sensor and by imaging, against the report's requirement and its test image (c).

mean NIS is 1.004 against [0.983, 1.017] over 26561 updates. The test also has power: $R \times 0.1$ gives an NIS of 6.8, $R \times 10$ gives 0.13, and $Q \times 0.01$ gives 1.68 (Fig. 9).

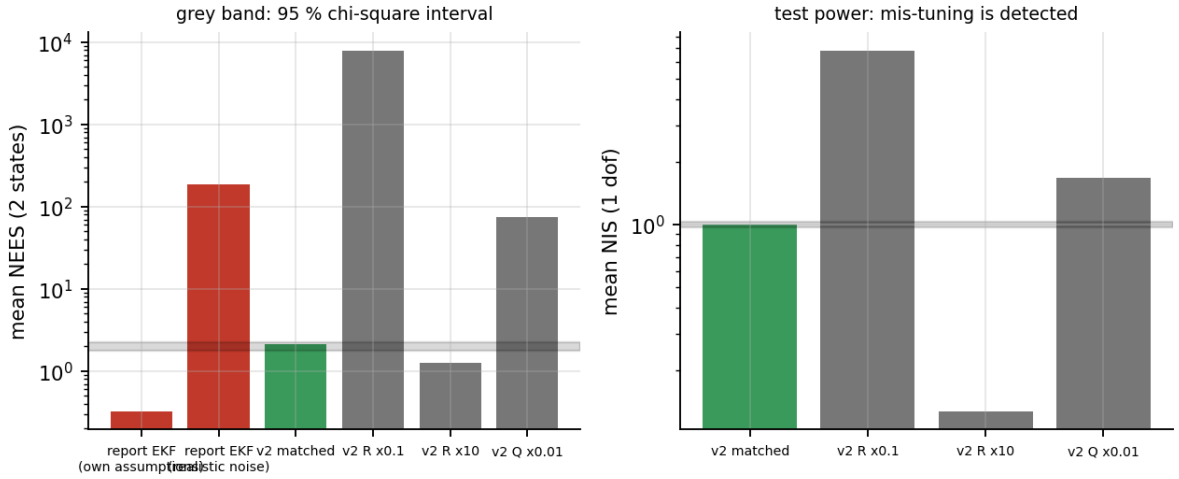


Figure 9: Mean NEES and NIS for the report's and the redesigned filters. The report's filter fails even under its own assumptions; the redesigned filter lies near the bounds, and deliberate mis-tuning is detected.

6.3 Ablations and Joint Calibration

Figure 10 shows the estimator ablation. The median RMS attitude error over a 90 s sweep is 0.102 [0.098, 0.110] deg for the full filter. Without a bias state it is 0.76 [0.71, 0.91], even with the report's rest-calibrated gyro (7× worse), and 25.8 [21.4, 30.1] with the datasheet zero-rate offset. It is 5.5 [5.5, 5.5] when the 0.3 s latency is ignored, 0.100 [0.095, 0.106] with the star tracker only, 1.11 [1.00, 1.18] with the sun sensor only, and 5 [4, 6] for the as-built optics. Pure gyro integration is useless within the sweep (100 [92, 104]).

On the real Fig. 37 samples (n =21 paired points before the star-tracker misidentification; Fig. 11), the raw ST–SS discrepancy is 5.0° RMS. A gain/offset calibration of the sun sensor against the star tracker reduces it to 1.5° in leave-one-out cross-validation (95% bootstrap [0.9, 1.9]). This is a joint use of the heat pad and the star field that needs no new sensors.

7 Star Identification on the Real Star Field

Map. The 56 star pixel coordinates in the team repository were converted to (ϕ, θ) using the repository's own constants (a 102°×67° reference image, 4608×2592 px). The field spans $-65° \ldots 7°$ in ϕ but only 10° in θ (Fig. 12). The committed `star_table.npy` (46 rows) cannot be reconciled with the CSV (56 rows, 6 removed → 50), so the documented construction was used.

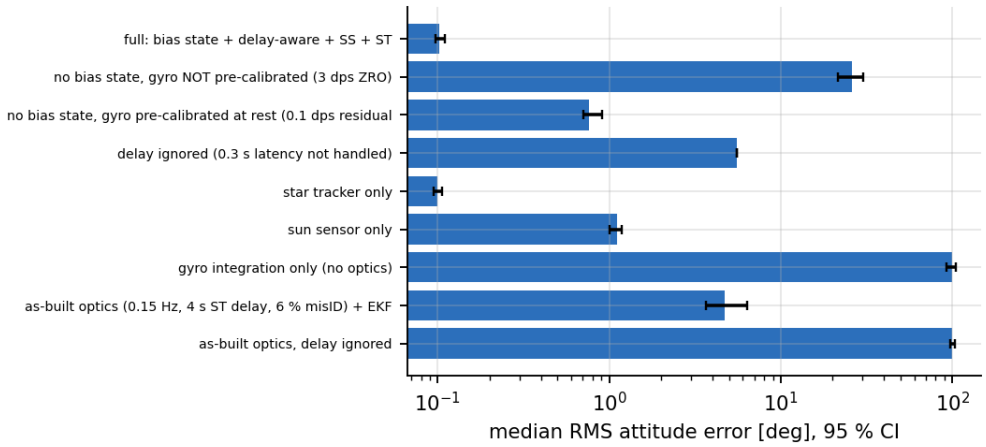


Figure 10: Estimator ablation: median RMS attitude error with 95% CI (log axis) for each filter variant.

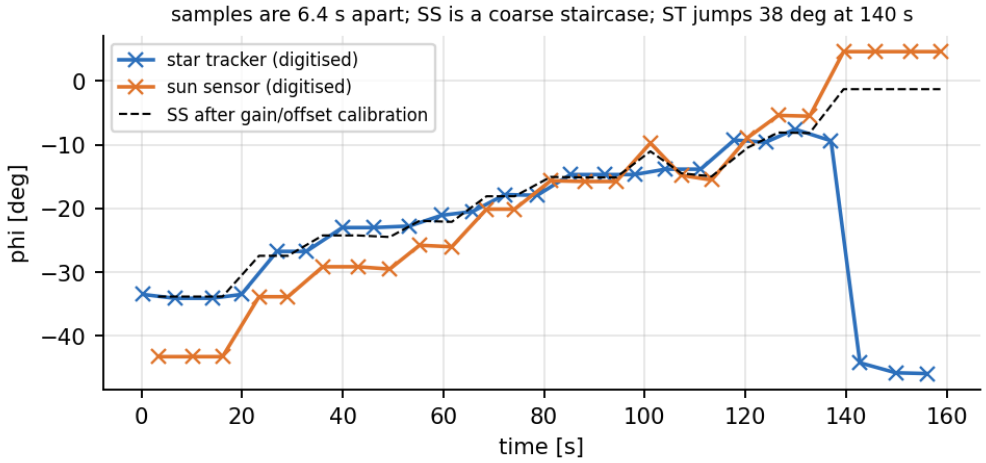


Figure 11: Re-analysis of the report's Fig. 37: samples every 6.4 s, the staircase sun sensor, the 38° star-tracker jump, and the gain/offset-calibrated sun sensor.

Degeneracy. At the report's tolerance of 0.004 rad, 99% of all star-pair distances have a coincident partner, and 92% of stars share their first-neighbour distance with another star (3.4 extra candidates on average; median nearest-neighbour distance 0.028 rad). The unique match therefore rests on the later '≥ 5 distance matches' check, which behaves correctly on complete, noise-free data (100% self-match).

Simulation. Images are rendered from the real map with 0.5 px centroid noise, a catalogue error of 0.002 rad, 90% detection and 0.5 false stars per image. The report's matcher (a line-by-line port of `signed_match_to_lookup`) returns the wrong star in 4.9% [3.6, 6.6] of images with the code's 30° crop and 5.1% [3.7, 6.8] with the report's 20° crop (no result 7.6% [6.0, 9.7]), and is correct in 93.6% [91.7, 95.1] and 87.3% [84.8, 89.5] respectively. In pose voting, each (observed, catalogue) pair votes for an offset, and the result is accepted if the peak has ≥ 4 votes and beats every peak at a different ϕ by 2. Pose voting gives 100.0% [99.5, 100.0] and 99.9% [99.3, 100.0] correct with 0.0% [0.0, 0.5] wrong (Table 5, Fig. 13). It tolerates 60% detection (100% against 64%) and a 0.004 rad catalogue error (88% correct and never a wrong star, against 48% correct and 22% wrong for the report). It exploits the one-axis constraint (a pure translation), so it is a special case of geometric voting [10]. As for *speed*, in this port the matcher core takes ≈0.56 ms and pose voting 2.80 ms per image, so the report's 3–5 s is not spent in the matching. Additionally, a triangle area is a function of its three side lengths, so 'area plus distance' adds an index, not information [11].

Field design (dropped). A simulated-annealing design of the 56 positions under coverage and spacing constraints lowers the difference-vector ambiguity cost from 546 (random) and 783 (real field) to 90. However, the identification yield under stress (12° crop, 70% detection, 1 false star/image, 0.003 rad map error) is 62.7% [58.8, 66.6] for random fields with the same spacing (mean of eight), 54.4% [49.4, 59.3] for the real field and 64.6% [61.4, 68.4] for the optimised fields (mean of eight). The annealed layouts beat the *real* field by roughly ten points, but random fields with the same spacing already reach about the same yield, so the gain comes

from uniform coverage (stars per window) and not from the uniqueness criterion. Once pose voting is used, the yield is limited by the stars per window and the detection probability. The uniqueness-design idea is therefore dropped, and the practical advice is to space the stars evenly and add more of them where the real field has gaps.

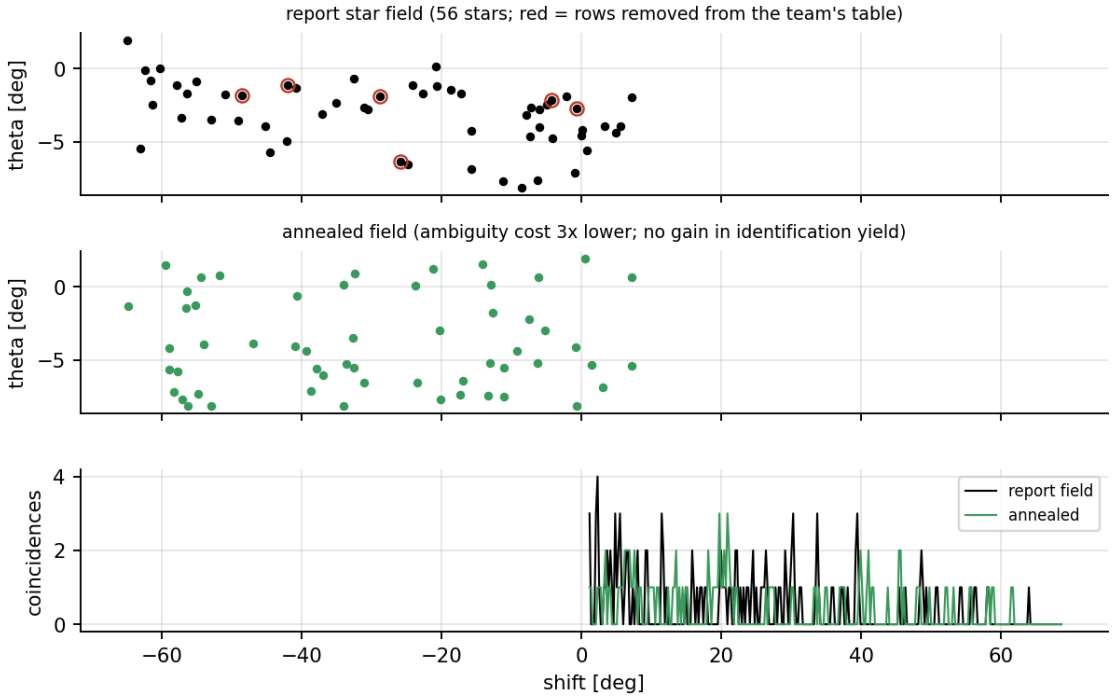


Figure 12: Star-field layouts and their ambiguity. Top: real field. Middle: annealed field. Bottom: shift-coincidence counts at 0.004 rad tolerance.

Table 5: Star identification on the report's real star map (simulated images: 0.5 px centroid noise, 0.002 rad catalogue error, 90% detection, 0.5 false stars/image). Wilson 95% intervals.

Crop	Algorithm	correct	wrong star	no result	ms/image
30deg_crop(code)	report	93.6% [91.7, 95.1]	4.9% [3.6, 6.6]	1.5% [0.9, 2.6]	0.56
30deg_crop(code)	hough	100.0% [99.5, 100.0]	0.0% [0.0, 0.5]	0.0% [0.0, 0.5]	2.80
30deg_crop(code)	pairvote	100.0% [98.1, 100.0]	0.0% [0.0, 1.9]	0.0% [0.0, 1.9]	8.49
20deg_crop(report text)	report	87.3% [84.8, 89.5]	5.1% [3.7, 6.8]	7.6% [6.0, 9.7]	0.65
20deg_crop(report text)	hough	99.9% [99.3, 100.0]	0.0% [0.0, 0.5]	0.1% [0.0, 0.7]	2.15
20deg_crop(report text)	pairvote	99.5% [97.2, 99.9]	0.0% [0.0, 1.9]	0.5% [0.1, 2.8]	4.33

8 Magnetic Subsystem and Test Coil

Rods. For a rod of diameter d and length l with N turns of 30 AWG (0.338 Ω/m), the dipole is

$$m = \mu_{eff} N I A, \qquad \mu_{eff} = \frac{\mu_r}{1 + N_d(\mu_r - 1)}, \tag{6}$$

where A is the rod cross-section, N_d is the prolate-spheroid demagnetising factor, and $\mu_r = 200$ is assumed. The 493-turn rod is 3.4 Ω and draws 1.47 A, giving $m \approx 1.14$ A m^2 (the report measured 0.95 by a position-fitting method). A 1500-turn, 80 mm redesign gives 1.02 A m^2 at 0.43 A and 2.2 W. Since m scales with μ_{eff} and μ_r is unverified, the moment must be measured.

Drive electronics. A bipolar H-bridge (MOSFET or integrated driver) with current sense, hardware PWM at $\geq$ 20 kHz and a flyback/TVS path replaces the relays. At 2 W per torquer and 0.4 A for the 1500-turn rod this is a simple circuit, whereas the old relay design could not give proportional or reversed dipoles.

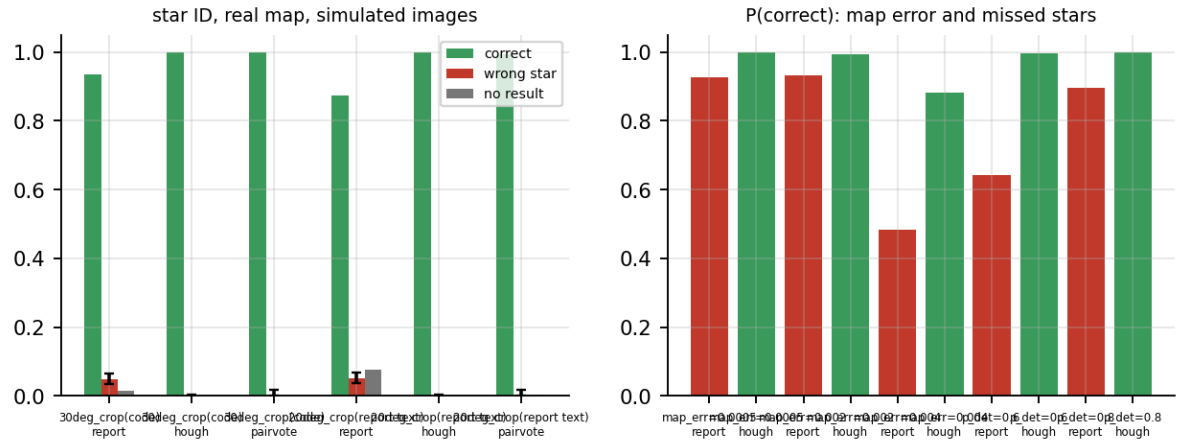


Figure 13: Star-ID outcome rates on the real map, and their sensitivity to catalogue error and missed stars.

Test coil. A Biot–Savart summation over finite straight segments for a square pair gives an optimum separation of $0.546\,a$ (literature 0.5445), where a is the side length. For the built 12 in coils at 14 in, the field varies by 26% p-p over the 10×10×15 cm body. A 0.6 m pair at 0.327 m gives 0.6% and needs 55 A-turn for 150 μT (e.g. 25 turns of AWG 18 at 2.2 A, 5.6 V, 12 W). The report's 'N cancels' argument holds only for a voltage source.

Torque authority and de-saturation. With the identified friction (4.1×10^{-4} N m), the available magnetic torque (mB, with 0.95–1.6 A m^2 at 90 μT) is only 0.21–0.35 of the friction. The testbed therefore cannot show magnetic detumbling at all, whereas the same torquers exceed an air-bearing residual by 14×. Meeting Requirement A by magnetic torque alone in 60 s would need 2.1 A m^2 at 90 μT.

Momentum management (S4). On an air bearing (10^{-5} N m friction, with the field at 40–100% of nominal and a 10° direction error to represent Table 10's gradient and misalignment), a wheel at $h = 0.02$ N ms must be dumped. The open-loop concurrent S4 of the report's concept loses pointing in 100% of trials. Closed-loop attitude control plus magnetorquer momentum feedback keeps the pointing within 0.19° rms and reaches 20% of $|h|$ in 325 [307, 335] s (68% within 500 s), close to the bound $H/(mB) \approx 317$ s. A coordinated LQ design reaches the same time (326 [307, 335] s) with a worse attitude transient (6.3°). Coordination therefore gives no dump-time benefit, because the limit is mB; it is dropped as an improvement (§10). Figure 14 summarises the magnetic results.

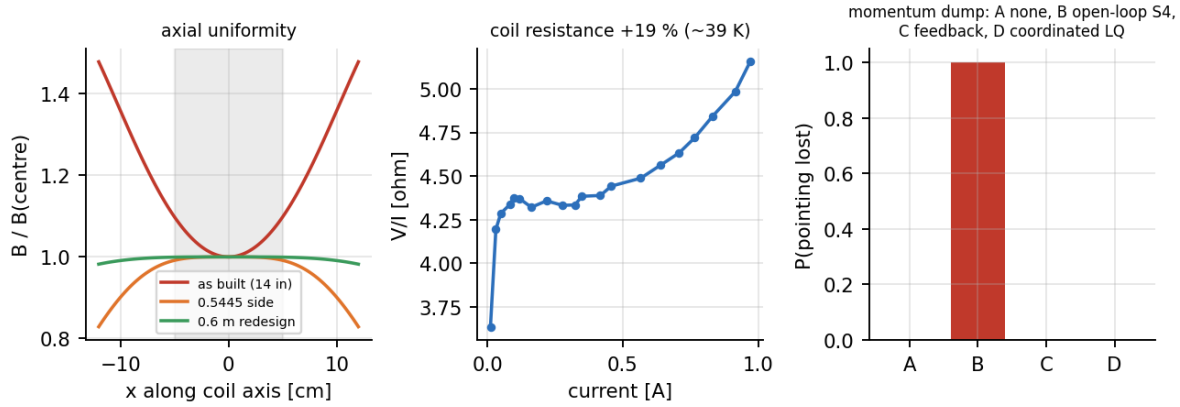


Figure 14: Magnetic subsystem results. Left: axial coil uniformity (shaded: body extent). Middle: measured coil resistance rise (Table 10 of the report). Right: probability of losing pointing during the momentum dump.

9 Imaging

The optics have an IFOV of 62 μrad, a GSD of 0.31 mm at 5 m and an FOV of 14.3° × 10.8°. The 25 cm sphere spans 806 px and a 1 mm feature is 3.2 px. The diffraction limit is an Airy radius of 1.04 px at f/2.4,

and the total depth of field is 0.59 m at 5 m (2-px circle of confusion). A 0.7 mm resolved feature is 2.3 px, so the report's static resolution claim is consistent. The 60 fps video, however, needs a binned mode on this sensor (unverified), in which 1 mm is 1.6 px, below Nyquist.

The smear MTF is $\mathrm{sinc}(fs)$ at spatial frequency f. At the 1 mm line-pair frequency, MTF= 0.5 at $s = 3.9$ px, so the 0.5 px budget is conservative. To test this, we render a crater scene (1 mm craters, Poisson noise plus 3 e^- read noise), apply exact uniform motion blur, and compare (i) a 10 ms exposure, (ii) an exposure shortened to 0.5 px of smear, (iii) K gyro-registered short frames (registration error 0.15 px, growing with $\sqrt{1 + k/10}$, with K limited by the dwell time in the 14.3° field), and (iv) a strobe 20× brighter. The fidelity ρ (correlation with the sharp scene) and the crater detection at 1% false alarm give the maximum rate with $\rho \geq 0.8$ and $P_d \geq 0.9$. At 2×10^5 e^-/px/s, the maximum tested rate meeting the criterion is none for the 10 ms exposure (rad/s, 'none' = never), 0.05 for stacking and 0.02 for strobing; at 8×10^5 it is 0.02 (naive) and 0.2 (stacked) (Fig. 15). Stacking and strobing therefore matter only if the rate cannot be held below ∼6 mrad/s, which the hold controller achieves.

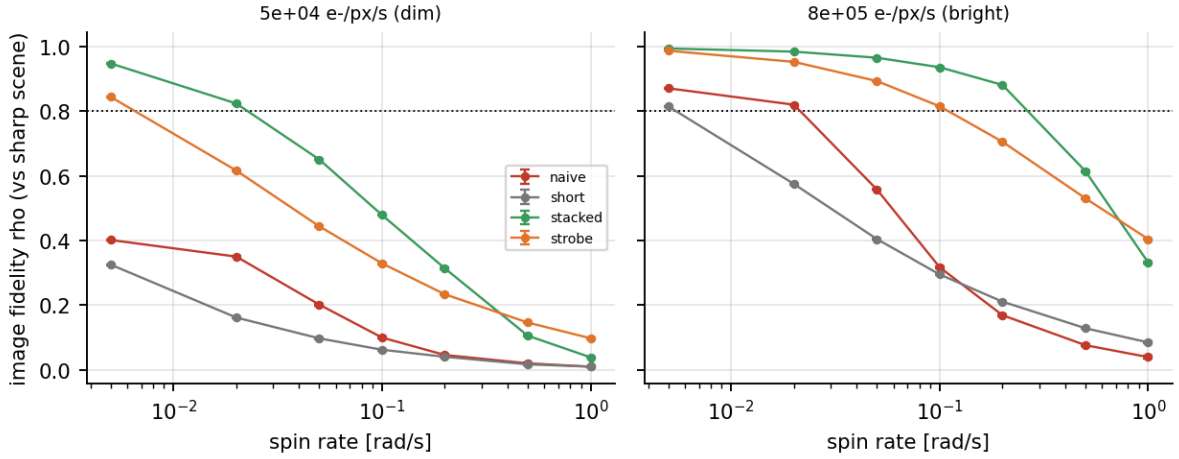


Figure 15: Image fidelity versus spin rate for four imaging strategies at two illumination levels (95% CI over 60 scenes).

10 Novel-Methods Workstream

Candidate improvements were evaluated against the redesigned baseline by deterministic Monte Carlo with confidence intervals, and screened against the literature (Table 6). The verdicts use three terms: HOLDS-REPLICATED (the gain is real, but the idea is already published), HOLDS-APPLIED (the gain is real and the idea is standard, so the value is in the quantified application), and DROPPED.

N1 Time-optimal manoeuvres. These give faster rest-to-rest slews, 13% [11, 14] at 90° and 10% [9, 11] at 150° against the PID (§5), but no detumble gain, since torque limits every controller. The method is established [7, 8]. **N2 Wheel+MTQ.** Dropped (§8). **N3 Fusion.** The bias state, delay-aware updates and SS–ST calibration give the large gains of §6; all are standard [12, 13, 14]. **N4 Star ID.** Pose voting removes wrong IDs in simulation; the speed claim is dropped. **N5 Field design.** Dropped. **N6 Smear defeat.** Known [17, 18]; the operating envelope is quantified here. **N7 Testbed.** Fitting pulse-time series instead of three stop times shrinks the friction uncertainty from -32/+71% to ±0.7% (single Hall magnet) or ±0.001% (360 cpr) in simulation, with the caveat that real bearings add Stribeck and stick-slip model error. An air bearing (residual 10^{-6}–10^{-5} N m [9]) is required to test the magnetics, and it must be levelled to 0.020° for a 3 mm CoM offset.

11 v2 Baseline and Test Campaign

Architecture. The v2 baseline is a 2U spacecraft (100×100×227 mm) with the same sensor suite, a wheel with an encoder and closed speed loop, H-bridge torquers, a 4S 1300 mAh pack, a ±1000 dps gyro range, a strobe-capable target light, and an air-bearing platform. The 1U option cannot route three cameras and an 80 mm wheel (component volume 553 cm^3; 2U fill 24%). The mass is 858 g (987 g with 15% margin; < 1 kg).

Table 6: Novel-methods workstream: verdicts against the literature (S-numbers in `research/sources.md`).

ID	Idea	Verdict	Literature / why
N1	time-optimal (bang-bang with torque saturation) detumble and slew	HOLDS-REPLICATED	Time-optimal attitude manoeuvres with wheel limits are established (S25-S29, incl. flight results >10 % shorter slews)
N2	reaction-wheel + magnetorquer momentum management in one optimal scheme	DROPPED (as an improvement)	Optimal wheel/magnetorquer momentum unloading is established (S20-S22: Camillo & Markley 1980, Chen et al. 1999, Chen et al. 2010)
N3	sensor fusion: gyro-bias estimation from optical fixes; delayed/multi-rate updates; joint heat-pad + star-field calibration	HOLDS-REPLICATED	Gyro bias from star-tracker data (S52, S53), delayed/out-of-sequence updates (S59, S60), MLX90640 sun sensing with MEKF (S50, S51), MEKF (S57, S58)
N4	faster / more robust star identification (pose voting exploiting the 1-axis constraint)	HOLDS-APPLIED	Geometric voting (S46), planar-triangle ID (S45), pyramid (S42); a translation Hough vote is a special case
N5	star-field design for uniqueness (difference-vector multiplicity annealing)	DROPPED (uniqueness criterion)	no direct star-field-design-for-uniqueness paper found in the survey (S48-S49 optimise catalogue databases); Golomb/Costas-type ideas exist in other fields
N6	exposure / shutter synchronisation, gyro-registered stacking, strobed illumination	HOLDS-REPLICATED	Gyro-aided deblurring (S61), MEMS-gyro restoration of star images (S62), attitude-correlated frames (S63), rolling-shutter compensation (S64-S66)
N7	testbed: air bearing and coast-down friction identification	HOLDS-APPLIED	Air-bearing simulators (S30-S32), low-cost testbeds (S33-S36), friction identification (S37, S38)

The energy for the 21 min campaign is 6.5 Wh against 19.2 Wh (margin 2.4×), and the peak current is ≈2.1 A against the report's 2 A assumption. All loads are assumed and will be replaced by test T9, and the battery energy density of 140 Wh/kg is derived from verified pack data. The requirements, mass and power budgets are given in Tables 7, 8 and 9.

Table 7: Re-baselined requirements (verification and acceptance rules).

ID	Requirement	Verification / acceptance	Repairs
R-DET-1	From a spin of 1.0 rev/s (range 0.5-1.7 rev/s), the ADCS shall reduce the rate below 10 deg/s within 60 s	Test on air bearing (residual <1e-5 N m), ADCS-on vs ADCS-off paired runs; N>=20 runs, >=95 % pass, one-sided 95 % lower Wilson bound >=85 %; ADCS-off runs must fail	A03, A04
R-DET-2	From 10 deg/s, reduce to <= 0.5 deg/s within 60 s	Test on air bearing; as R-DET-1	A03
R-ACQ-1	Acquire attitude (EKF initialised, sigma < 1 deg) within 30 s of reaching 0.3 rad/s from unknown attitude	Test with randomised start angle (>=20 runs); median <= 20 s, p95 <= 30 s	A24
R-PNT-1	Hold the target within +/-3.5 deg (target sphere inside the 10.8 deg short axis) with rate <= 6.2 mrad/s for >= 10 s (stretch: 60 s)	Test; attitude from encoder/table reference; fraction ready >= 0.95 over 3 x 20 runs	A07, A08, A23
R-IMG-1	Image 1 mm features at 5 m: GSD <= 0.33 mm (k=3), smear <= 0.5 px, static MTF50 >= 1 mm line-pair frequency	Slanted-edge MTF at operating exposure and mode; image of calibration chart at 5 m; MTF at 0.155 cyc/px >= 0.4; chart groups to 0.7 mm resolved in 3/3 frames	A19, A08

ID	Requirement	Verification / acceptance	Repairs
R-IMG-2	Exposure/illumination shall deliver >= 20 dB SNR at the chosen exposure without saturation	Photometric test with strobe/continuous lamp; SNR measured on grey patch >= 20 dB at <= 5 ms	A07
R-ADS-1	EKF attitude error <= 0.1 deg (1-sigma) during hold, NEES and NIS consistent with chi-square 95 % intervals over >=500 simulated and 5 hardware runs	Analysis + replay of logged data; time-averaged NEES within the interval or conservative; NIS mean within [0.97,1.03]	A21
R-ADS-2	Gyro range >= 1000 dps (+/-17.5 rad/s) with saturation flag logged; bias estimated online	Inspection + test with 1.7 rev/s spin; no saturation flag in 20 runs; bias estimate converges within 10 s	A04
R-STR-1	Star identification: P(correct) >= 0.99, P(wrong) <= 0.5 % per attempt over >=1000 stored images; processing <= 1 s on the Pi	Test on stored images + analysis; Wilson upper bound on P(wrong) <= 1 %	A22
R-SUN-1	Sun sensor update >= 1 Hz (full frame at refresh >= 2x), residual vs encoder <= 0.5 deg after gain/offset calibration	Rotary table with encoder; RMS <= 0.5 deg over 3 sweeps	A20
R-DMP-1	Reduce wheel momentum by 0.02 N m s (10.7 rad/s-equivalent) within 400 s while holding attitude within 1 deg	Test on air bearing, B >= 90 uT, m >= 1.0 A m^2; N>=10 runs, >=90 % reach 20 % of initial |h| within 400 s	A13
R-MTQ-1	Torquer pair: dipole >= 1.0 A m^2 each at <= 6 W, bipolar drive via H-bridge, flyback protected, thermal cut-off at 80 C	Helmholtz-cage moment measurement with uncertainty budget; m within +/-15 % of design in 3 runs	A09, A10, A11
R-WHL-1	Wheel: torque >= 1 mN m, momentum capacity >= 0.04 N m s, speed measured to 0.1 RPM, command resolution <= 0.2 RPM (16 bit)	Spin-up/coast-down with encoder; resolution test: 0.2 RPM step visible in encoder	A02
R-PWR-1	Operate >= 15 min with >= 2x energy margin and a logged power profile; peak current <= 4 A; no brown-out of the 5 V rail	Bench + full-system run with logger; min 5 V rail >= 4.75 V in all runs	A18
R-MAS-1	Mass <= 1.0 kg with 15 % margin; envelope 100 x 100 x 227 mm (2U)	Scale + inspection; mass measured <= 1.0 kg	A25
R-TB-1	Testbed residual disturbance torque <= 1e-5 N m (air bearing), levelled to <= 0.02 deg for a 3 mm CoM offset	Coast-down with 360+ cpr encoder, fit Coulomb+viscous; Coulomb torque estimate 95 % CI upper bound <= 1e-5 N m	A03, A16, A17
R-TB-2	Magnetic test field >= 3x Earth (90 uT) uniform to <= 10 % p-p over the 10 x 10 x 15 cm body, reproducible to 10 %	3-axis grid map at 3 currents, 3 days; p-p <= 10 %, day-to-day <= 10 %	A14, A15

Test campaign. Table 10 lists ten tests with their sample sizes and acceptance rules. Every pass/fail rule uses a Wilson lower bound or a CI, and every dynamic test pairs ADCS-on with ADCS-off runs. The schedule (Table 11) is 14 weeks with a margin week.

12 Comparison with Current Practice

The results so far judge the testbed against its own requirements. To place it in context, we now compare its sensors, its achieved and required accuracy, and the testbed itself with commercial CubeSat ADCS hardware,

Table 8: v2 mass budget (g). Status: DERIVED from CAD/verified data or ASSUMED catalogue value.

Item	Mass	Status
structure (bottom plate PETG + 3 Al plates + brass stand-offs, 2U frame)	250	DERIVED from CAD (cad_inertia) scaled to 227 mm
reaction wheel + BLDC motor + encoder + mount	150	ASSUMED
2 magnetorquers (1500 turn, 80 mm rod)	68	DERIVED (a5)
PDB with H-bridge drivers	45	ASSUMED
Raspberry Pi 5	46	ASSUMED (catalogue)
Arduino-class MCU	10	ASSUMED
IMU (ICM-20948) x2 + breakout	8	ASSUMED
HQ camera + 25 mm lens	62	ASSUMED
star camera (Module 3 wide)	8	ASSUMED
MLX90640 sun sensor	4	ASSUMED
battery 4S 1300 mAh	137	DERIVED from verified energy density
harness, fasteners, brackets	70	ASSUMED
total	858	+15% margin = 987

Table 9: v2 power budget by mode (ASSUMED loads, 87% converter efficiency; replaced by measurement in test T9).

Mode	load [W]	battery [W]	current at 14.8 V [A]
detumble (wheel high torque)	26.9	30.9	2.09
capture / slew	16.9	19.4	1.31
imaging hold	10.9	12.5	0.84
S4 dump	15.3	17.6	1.19

with flown 1U–6U missions and with university air-bearing facilities. The comparison uses published numbers only (`research/sources_v3.md`); where a value is a vendor or aggregator snippet rather than a document read in full, it is marked in that file.

12.1 Integrated ADCS Units and Flown Missions

Table 12 lists the integrated units and components that a flight version of this design would be built from. The NASA small-spacecraft state-of-the-art survey gives a range of 0.002–5° for integrated units, 0.15–15 A m^2 for magnetorquers and 0.0005–8 N ms for wheels [20]. A complete three-axis unit with a star tracker now weighs 0.26–1.7 kg, which is comparable to the whole 858 g v2 testbed spacecraft.

Figure 16 places the published pointing performance of flown missions and commercial units by actuator class. Passive magnetic stabilisation holds 15–18° to the field (CSSWE, BEESAT-3) [28, 29], magnetorquer-only control gave 10° on GOMX-1, wheels with torquers give about 0.1–2° (KU Leuven ADCS, GOMX-3) [30, 31], and star-tracker systems reach 15–42″ (3σ) on the 3U MinXSS-1 [26] and 0.5″ rms on the 6U ASTERIA [27]. The bases differ (1σ, 3σ, rms), so the chart orders classes rather than ranking products. Against this scale the testbed's numbers are those of the coarsest class. The as-built star-tracker/sun-sensor discrepancy of 5.0° rms is worse than a magnetorquer-only flight system, the v2 pointing requirement R-PNT-1 (±3.5°) sits between the magnetorquer-only and wheel classes, and only the v2 estimator requirement R-ADS-1 (0.1°) reaches the level of current wheel-based units. This directly shows that the testbed is a training and verification tool, not a precursor of a competitive flight ADCS, and the requirements of Section 11 should be read in that light.

12.2 Star Trackers and Sun Sensors on Small Satellites

Figure 17 compares the testbed's optical sensors with flight units. Commercial small-satellite trackers are accurate to 6–72″ cross-axis but are rate-limited: CubeStar tracks to 0.3°/s and the BCT Nano Star Tracker to 2°/s, and a published survey notes that accuracy at rate is not well characterised for most trackers [20, 22, 23]. The testbed's star camera tolerates a higher rate (0.058 rad/s, about 3.3°/s, set by blur on a printed field 5 m away) only because its targets are bright and its accuracy requirement is coarse, so the two are not comparable devices. The sun sensor is the clearer gap: the measured 0.16 Hz update and 5.0° rms error sit two orders of magnitude in rate and one in accuracy from a 5 g commercial sensor (>10 Hz, < 0.5°

Table 10: v2 test campaign.

ID	Test	Objective	N	Acceptance
T1	Static optical characterisation	GSD, MTF, focus, mode, exposure (R-IMG-1/2)	3 sessions	slanted-edge MTF, 5 m chart
T2	Gyro/IMU calibration	six-position, Allan variance, scale, range (R-ADS-2)	12 h static + rotary table	bias instability and ARW published
T3	Sun sensor and star tracker calibration	angle vs rotary-table encoder (R-SUN-1, R-STR-1)	>=1000 images	RMS and P(wrong)
T4	Friction/air-bearing characterisation	coast-down fit (R-TB-1)	5 runs x 3 speeds	Tc upper bound <= 1e-5 N m
T5	Wheel characterisation	torque, resolution, momentum, coast (R-WHL-1)	10 runs	resolution, time constants
T6	Magnetorquer and Helmholtz mapping	dipole, field uniformity (R-MTQ-1, R-TB-2)	3 days	m within 15 %, p-p <= 10 %
T7	Detumble campaign	R-DET-1/2 paired ADCS on/off	>=20 runs each	Wilson bounds
T8	Acquire, point, image	R-ACQ-1, R-PNT-1, R-IMG-1	>=60 runs	ready fraction; image chart groups
T9	Power and endurance	R-PWR-1	3 x 15 min	rail min voltage
T10	Momentum dump	R-DMP-1	>=10 runs	time to 20 % of |h|

Table 11: Fourteen-week schedule.

Weeks	Requirements	Work
W1-2	R-*	Re-baseline requirements, order long-lead items (encoder wheel, air bearing, H-bridge, strobe)
W3-4	R-IMG-1, R-ADS-2, R-SUN-1	Bench: IMU/gyro, optics MTF, sun sensor and star tracker calibration (T1-T3)
W5-6	R-TB-1, R-TB-2	Testbed: air-bearing platform + encoder, friction coast-down (T4); Helmholtz rebuild (T6)
W7-8	R-WHL-1, R-MTQ-1	Wheel and torquer characterisation (T5, T6); PDB v2 with drivers
W9-10	R-ADS-1, R-STR-1	Software: EKF (bias, delay), pose-voting star ID, controller, S4; replay on logged data
W11-12	R-DET-*, R-PWR-1	Integration, power profile and endurance (T9), detumble campaign (T7)
W13	R-ACQ-1, R-PNT-1, R-DMP-1	Acquire/point/image campaign (T8), momentum dump (T10)
W14	-	Margin week, report

rms) [24], which is why R-SUN-1 asks for $\geq$1 Hz and 0.5°.

12.3 University Air-Bearing Testbeds and Helmholtz Cages

Figure 18 compares the testbed itself with published facilities. Balanced three-axis tables reach a residual disturbance torque below 5×10^{-5} N m with automatic sliding-mass balancing (Bologna) [32], and a bench target of about 10^{-5} N m is quoted for 1–3U wheel experiments whose control torques are 0.6–1 mN m [33]; manual counterweight balancing is reported as not better than 10^{-3} N m [32]. The identified bearing friction of the team's platform, 4.1×10^{-4} N m, is therefore 8–40× above a balanced university table and less than a factor of three below the wheel torque it is meant to reveal. The v2 requirement R-TB-1 (10^{-5} N m) is at the edge of what university tables report and should be treated as a goal with 5×10^{-5} N m as the threshold. Published Helmholtz cages are uniform to 2–5% over a CubeSat volume [34]; the team's coil varies by 26% over the body (model) and its measured field falls by 63% between sensors 39 mm apart, while the 0.6 m redesign of Section 8 reaches 0.6%.

13 From Testbed to Orbit

The testbed was never meant to fly, but its purpose is to prepare a flight ADCS, so the question SMAD asks of any ADCS design [19] must eventually be answered: what environment, what disturbances, what actuator

Table 12: Commercial CubeSat ADCS Hardware Used as the Reference for a Flight Version

Item	Mass	Key performance	Source
CubeSpace CubeADCS (3-axis)	0.26 kg	≈70″ (3σ), configuration dependent	NASA SOA Table 5-2 [20]
AAC iADCS-200 / iADCS-400	0.47 / 1.7 kg	$< 1°$	NASA SOA Table 5-2 [20]
BCT XACT-15 / XACT-50	0.885 / 1.23 kg	0.003/0.007°; 15 / 50 mN ms	NASA SOA [20]; BCT datasheet (snippet)
CubeSpace CW0057 wheel	0.115 kg	5.7 mN ms, 2.7 W	NASA SOA Table 5-3 [20]
Rocket Lab RW-0.01 wheel	0.122 kg	10 mN ms, ±1 mN m	datasheet [21]
CubeSpace CR0002 / ISIS iMTQ	16.5 g / 0.196 kg	0.2 A m^2 (0.49 / 1.2 W)	NASA SOA Table 5-4 [20]
CubeSpace CubeStar tracker	0.047 kg	0.02° cross-axis, 0.3°/s max rate	NASA SOA Table 5-5 [20, 22]
NewSpace NCSS-SA05 sun sensor	<5 g	$< 0.5°$ rms, >10 Hz	datasheet [24]

Prices published for the CubeSpace Gen-1 three-axis system start at USD 15,390 (base unit) and USD 43,500 (full system), and CubeStar at USD 17,600 [25] (vendor sheet, snippet).

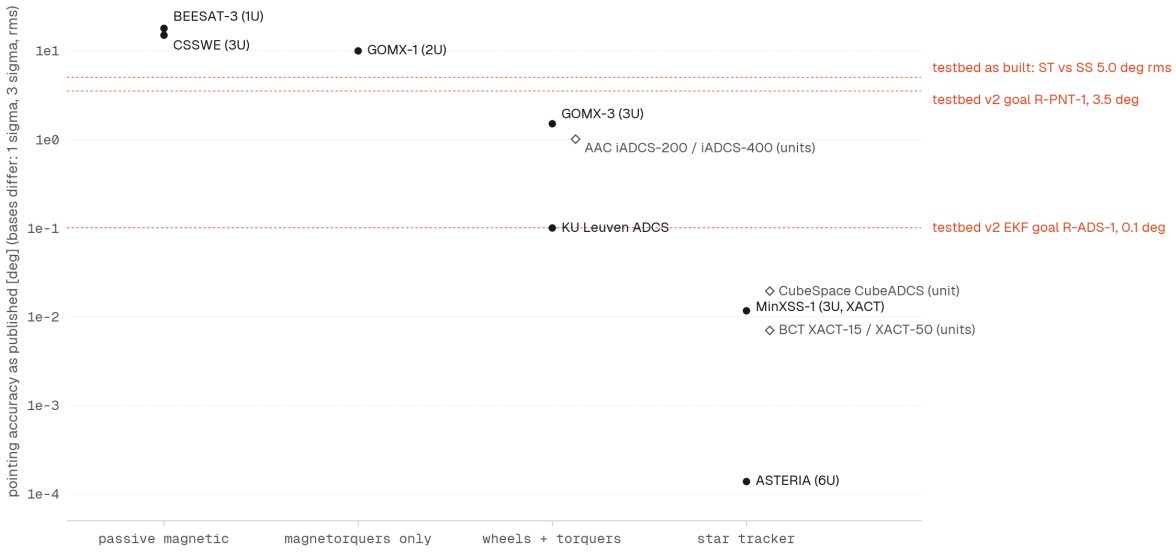


Figure 16: Published pointing accuracy of flown CubeSats (filled) and commercial ADCS units (hollow) by actuator class, with the testbed's as-built and v2 numbers (dashed). The testbed's as-built accuracy is that of the coarsest flight class. Sources in Table 12 and `research/sources_v3.md`; bases differ (1σ, 3σ, rms).

sizes, what power and data, and what verification. This section answers it for a flight version of the v2 2U (100×100×227 mm, 0.99 kg with margin, uniform-box inertia; a 2.0 kg case is also run because flight 2U spacecraft are usually heavier, and CubeSat Design Specification Rev. 14.1 allows 4.00 kg for a 2U [38]). The analysis is in `analysis/a10_flight.py`.

13.1 Orbit and Environment

Two orbits are considered: an ISS-deployment orbit at 400 km and 51.6° (deployment altitudes 380–420 km [37]) and a rideshare sun-synchronous orbit at 500 km (ASSUMED). The periods are 92.6 and 94.6 min and the longest eclipses are 36.1 and 35.8 min (cylindrical shadow at $\beta = 0$). The geomagnetic field is taken from the dipole model of SMAD p. 366,

$$B = \frac{M}{R^3}\sqrt{1 + 3\sin^2\lambda}, \tag{7}$$

where $M = 7.96 \times 10^{15}$ T m^3, R is the orbit radius and λ the magnetic latitude. This gives 25.6–43.1 μT at 400 km and 24.5–48.6 μT at 500 km. As a check, at 400 km and 60° the dipole gives a field 6.7% above the IGRF-14 value of 43,172 nT [35], which is adequate for sizing. The orbit field is therefore 2–4× weaker than the 90 μT test field of the report, which matters for every magnetic result below.

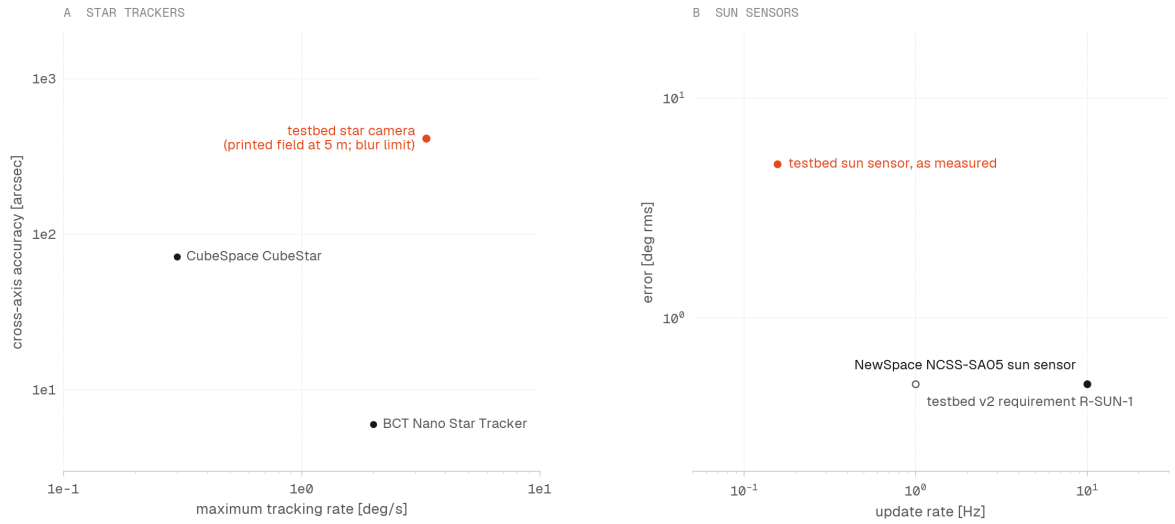


Figure 17: (a) Cross-axis accuracy versus maximum tracking rate for small-satellite star trackers and the testbed's star camera; (b) error versus update rate for a commercial sun sensor, the testbed's sun sensor as measured and the v2 requirement. The testbed's sun sensor is the largest sensor gap to flight practice. Sources: NASA SOA Table 5-5 [20], vendor data [22, 23, 24]; testbed values from Sections 5 and 6.

13.2 Disturbance Torques

We compute the worst-case external torques with the equations of SMAD Table 11-9A (pp. 366–367):

$$T_g = \frac{3\mu}{2R^3}|I_z - I_y|\sin 2\theta, \quad T_{sp} = \frac{F_s}{c}A_s(1+q)\cos i\,(c_{ps} - c_g), \quad T_m = DB, \quad T_a = \tfrac{1}{2}\rho C_d A V^2 (c_{pa} - c_g), \tag{8}$$

where μ is Earth's gravitational parameter, θ the deviation from local vertical (45°, worst case), $F_s = 1367$ W/m^2, $A_s = A$ the 0.1×0.227 m face, $q = 0.6$ and $C_d = 2.2$ (SMAD p. 367), D the residual dipole, ρ the density and V the orbital speed. The residual dipole is taken as 10^{-3} (ASSUMED), 10^{-2} (lower end of a published nanosatellite design value) and 5.85×10^{-2} A m^2 (the Aalto-2 estimate) [36]; the centre-of-pressure offset is 0.5 cm, 1 cm and 2 cm, the last being the CDS centre-of-gravity limit in X/Y [38]. No cited density table could be retrieved in this study, so the density is an exponential through the two values SMAD quotes on p. 367 (10^{-13} kg/m^3 at 700 km and 10^{-9} at 100 km), which over-estimates the density at 400–500 km and makes the aerodynamic torque an upper bound; the low case divides it by ten.

Figure 19 shows the result for the ISS orbit. The total ranges from 5.8×10^{-8} to 2.8×10^{-6} N m (nominal 5.9×10^{-7}), dominated by the residual dipole (2.5×10^{-6} N m in the high case) and then by drag (1.5×10^{-7} nominal); gravity gradient (6.6×10^{-9}) and solar pressure (1.7×10^{-9}) are negligible for this compact body. The sun-synchronous case is the same within 5% (2.9×10^{-6} high). Against the identified testbed friction, the orbit environment is 147–7,169× smaller. As a result of this, the testbed *over*-tests the disturbance torque by two to four orders of magnitude but cannot reproduce the slow, secular momentum build-up that sets wheel and torquer sizing in orbit.

13.3 Momentum Storage and Magnetorquer Sizing

Following SMAD Table 11-12 (pp. 370–371), the momentum accumulated per orbit is bounded by $T\,P$ if the torque never reverses, and is estimated as $T(P/4)(0.707)$ for a cyclic torque. In the high case this gives 1.6×10^{-2} and 2.8×10^{-3} N ms respectively (3.3×10^{-3} nominal, secular). A 5.7 mN ms CubeWheel would therefore saturate within one orbit in the secular worst case, and the 0.04 N ms of R-WHL-1 within about three. Momentum management is a flight requirement and not an option.

The dipole needed to cancel the disturbance at the weakest (equatorial) field, $D = T/B$ (SMAD p. 371), is 0.023 A m^2 nominal and 0.11 A m^2 in the high case (0.12 at 500 km). The commercial 0.2 A m^2 class (Table 12) is therefore sufficient for momentum management, with margin. For comparison, the 8.4 A m^2 implied by the report's de-saturation requirement S4 (A13) is 76× this. That figure follows from the requirement to dump 0.09 N ms in 120 s at 90 μT, not from any orbit environment, and is withdrawn as a sizing input for a flight design.

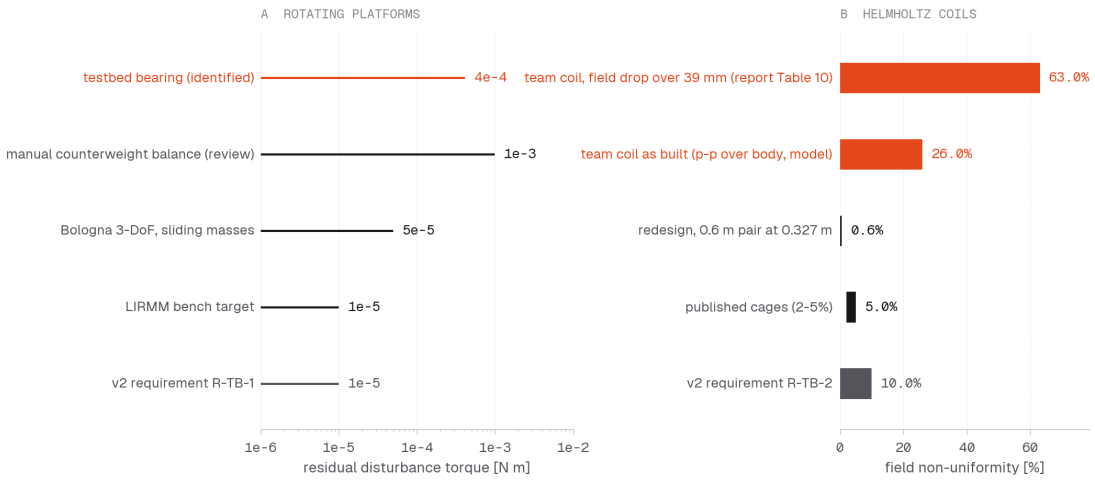


Figure 18: (a) Residual disturbance torque of the testbed bearing against university rotating platforms and the v2 requirement; (b) field non-uniformity of the team's coil against published cages and the redesign. The bearing is 8–40× worse than balanced university tables. Sources: [32, 33, 34]; testbed values from Sections 4 and 8.

13.4 Detumbling from Deployer Tip-off

Deployers release CubeSats with tip-off rates that are specified, not zero: the NanoRacks deployer targets $< 2°/s$ per axis and requires satellites to tolerate $5°/s$, the EXOpod specifies $< 10°/s$, and microgravity tests of a canisterised dispenser showed about $10°/s$ on the worst axis [40, 41, 42]. We therefore simulate a three-axis B-dot detumble in orbit. The rigid-body equations $J\dot{\omega} = -\omega \times J\omega + m \times B$ are integrated by RK4 at 1 s with quaternion attitude, the field is the axial dipole along a circular orbit, and the magnetometer is sampled and differenced once per step. The control law is a saturated proportional B-dot, $m = \mathrm{sat}(-k\dot{B})$, with one gain for all dipole classes, $k = J_{min}/(\frac{2}{3}B_{eq}^2\tau)$ and $\tau = 30$ s (ASSUMED), so that a larger rod only acts where a smaller one saturates. Three dipole classes are run (0.2 A m^2 commercial, 0.68 A m^2 as-built rod model, 1.14 A m^2 report rod model), from 2, 5 and $10°/s$ with 16 random initial attitudes and spin axes each.

Figure 20 shows the result. From $10°/s$ the 0.2 A m^2 class stays below $1°/s$ after a median of 1.1 orbits (90th percentile 2.0) at 400 km and 1.0 (1.5) at 500 km, and the rate then settles at $0.16°/s$, about twice the orbital rate, as expected for B-dot. The larger rods are not faster (median 1.1 and 1.4 orbits), and a 2.0 kg body with the 0.2 A m^2 class takes 1.6 orbits (90th percentile 2.0). A torque-limited bound, $J\omega_0/(m_{max}B\,2/\pi)$, is only 0.05 orbits for the 0.2 A m^2 class. This demonstrates that in-orbit detumbling is limited by the geometry of the field (the rate component along B cannot be damped until the field rotates, twice per orbit), not by the dipole above the commercial class. Oversizing has a cost instead: the largest rate change one 1-s sample can impose, $m_{max}B\Delta t/J_{min}$, is $0.36°/s$ for 0.2 A m^2, $2.0°/s$ for the 1.14 A m^2 rod and about $15°/s$ for an 8.4 A m^2 rod, so a strong torquer must be scaled down near the end of detumble.

13.5 Power and Data Budgets

SMAD builds the power budget mode by mode from the subsystem loads (Tables 10-8 and 10-9, pp. 314–317). For the flight 2U we take body-mounted triple-junction cells (28% efficiency, 50% face fill, 15% other losses: all ASSUMED). The orbit-average generation is then 2.7 W tumbling (mean projected area, one quarter of the surface for a convex body) and 3.3 W with the best fixed sun attitude, consistent with the < 10 W reported for body-mounted 3U spacecraft [43]. The loads use datasheet values where available (wheel, torquers, star tracker; Table 12) and ASSUMED values for the computer, radio and the report's Pi 5 payload (7 W). With the 87% conversion efficiency of the v2 budget, the battery-side power is 2.6 W in safe/detumble mode, 2.4 W in fine pointing, 6.5 W while downlinking and 10.5 W while imaging (Fig. 21).

The flight 2U can therefore point continuously, but imaging is power-limited. The energy balance closes with imaging for 10% of the time (about 9 min per orbit at 400 km; 11% at 500 km) after the downlink passes are paid for. Imaging through the longest eclipse discharges the v2 19.24 Wh pack by 33% (15% for a 43 Wh GomSpace BP4 [44]) against 8% in fine pointing. The Pi 5 payload, which the testbed budget treats as a continuous load, is thus the dominant flight power driver and must be duty-cycled or replaced by

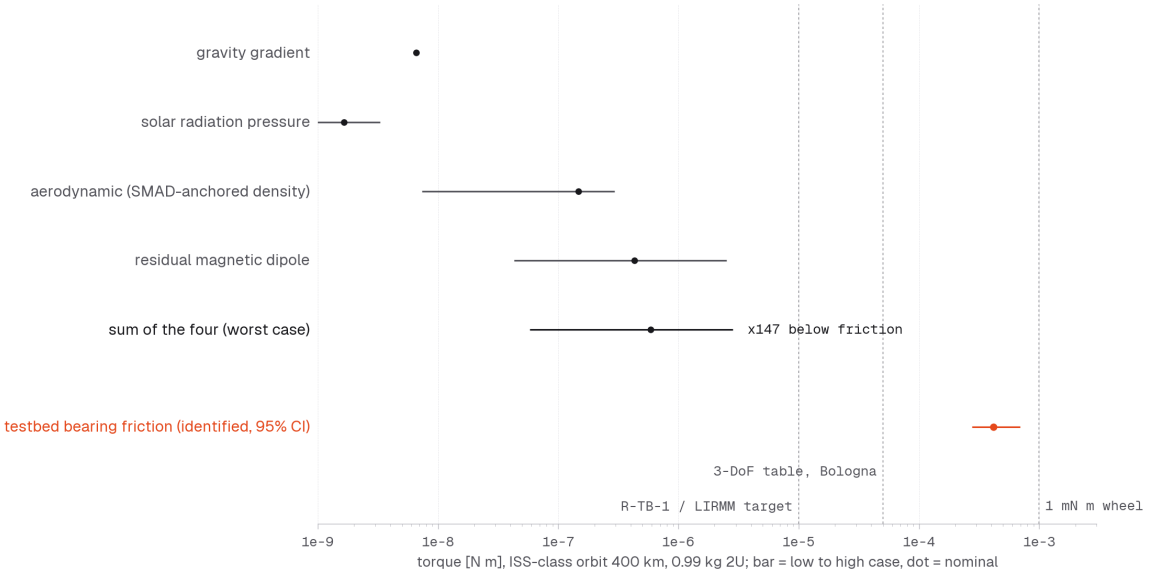


Figure 19: Worst-case disturbance torques on the flight v2 2U at 400 km (bar: low to high case; dot: nominal) against the identified testbed friction and published bench targets. Orbit disturbances are 147–7,169× below the testbed friction. Equations from SMAD Table 11-9A (pp. 366–367); inputs in the text.

a lower-power processor.

For the link to the ground we use the SMAD link equation (Eq. (13-13), pp. 554–555),

$$\frac{E_b}{N_0} = P + L_l + G_t + L_s + L_a + G_r + 228.6 - 10\log T_s - 10\log R, \tag{9}$$

with a 1 W UHF transmitter of the AX100 class at 9.6 kbps [45], a 0 dBi spacecraft antenna, a 12 dBi ground Yagi of the kind used by the SatNOGS network [46] (both ASSUMED), 3 dB of polarisation and pointing loss, $T_s = 221$ K for a UHF downlink (SMAD Table 13-10, pp. 558–559), and a required E_b/N_0 of 9.6 dB for BPSK at a BER of 10^{-5} plus 2 dB of implementation loss (SMAD Table 13-11, pp. 562–563). At 10° elevation (1695 km slant range at 500 km) the margin is 11.4 dB, and 4.8 dB if the urban noise temperature is 1000 K (ASSUMED). The link closes at every elevation above the mask (Fig. 22), so the data rate, not the margin, is the constraint.

A pass simulation for one mid-latitude station (Ann Arbor, ASSUMED; 10° mask; averaged over eight orbit planes and ten days) gives 5.7 passes and 30 min per day at 400 km and 51.6°, and 3.3 passes and 19 min at 500 km SSO. At 70% protocol efficiency (ASSUMED) the daily downlink is 1.52 MB and 0.96 MB respectively. A full-rate ADCS and housekeeping log (64 B at 1 Hz plus 2 B/s, ASSUMED) produces 5.7 MB/day, 3.8× the 400 km capacity, so flight telemetry must be decimated (0.73 MB/day at 0.1 Hz). What remains returns one 3 MB image every 3.8 days at 400 km and every 13 days at 500 km. The imaging objective of the testbed would therefore need an S-band downlink in flight; the ADCS objectives fit a UHF link once the log is decimated.

13.6 Verification Path, Cost and Reliability

Figure 23 sets out a path from the testbed to an orbital mission. The v2 testbed (air bearing, encoder wheel, H-bridge torquers) verifies the estimator, the star-identification logic and the controller structure. A three-axis FlatSat with hardware-in-the-loop, driven by an orbit simulation and a cage uniform to 2–5% [34] at the orbit field of 25–49 μT, is needed to verify B-dot and momentum management, because the one-axis platform cannot. Qualification then follows the launch provider's levels, with GEVS as the fallback: 14.1 grms qualification random vibration and about 10 grms acceptance [39], thermal-vacuum cycling and bakeout to the CDS outgassing limits (TML $\leq$1.0%, CVCM $\leq$0.1%) [38]. CDS Rev. 14.1 also imposes three RF inhibits and no transmission for 45 min after deployment [38], which the flight software must respect before the first detumble telemetry.

The results that transfer from the testbed to orbit are the estimator consistency tests, the star-ID voting

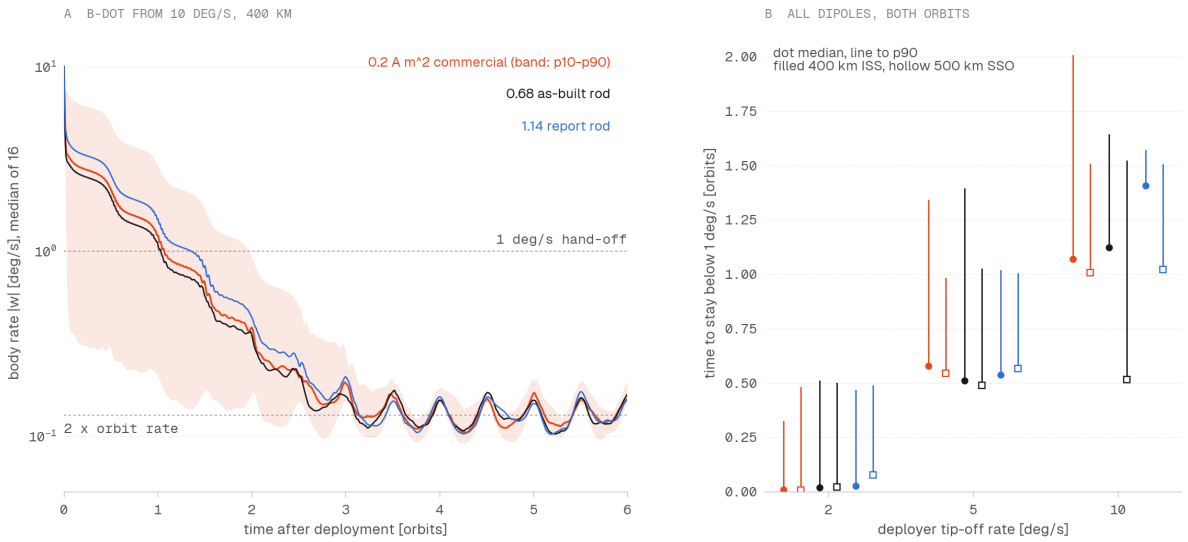


Figure 20: In-orbit B-dot detumble of the flight v2 2U. (a) Median body rate from 10°/s at 400 km for three dipole classes (band: 10–90th percentile, 0.2 A m^2). (b) Time to stay below 1°/s versus tip-off rate, both orbits. All classes detumble within about two orbits; the dipole above 0.2 A m^2 does not shorten it. Simulation in `analysis/a10_flight.py`; tip-off rates from [40, 41, 42].

logic, the controller structure and the units and requirements discipline of Section 16. The results that do not transfer are the one-axis dynamics, the friction-dominated detumble, the printed star field and the 90 μT test field.

For cost, a commercial three-axis unit starts at about USD 15,000 and a full system with a star tracker at about USD 43,500 [25]. Rideshare launch is quoted at USD 7,000/kg with a USD 350,000 minimum in 2026 [47], although NASA's CubeSat Launch Initiative can provide a launch at no cost to selected payloads, and one university's rough estimate for a 3U mission is USD 1M for the spacecraft, 0.25M for launch and 0.3M for operations [48]. For reliability, about a quarter of recent CubeSats were dead on arrival [49], and in a study of 178 CubeSats the power subsystem caused more than 40% of the failures after 30 days [50]. SMAD's observation that failure rate grows with complexity (Table 19-4, pp. 764–765) argues for buying the ADCS as a flight-proven unit and spending the student effort on the payload, operations and verification.

14 Recommendations for a v2 Flight Design

Based on the comparison and the flight sizing, the following recommendations apply to a flight version of the v2 design:

1. **Buy, do not build, the flight ADCS.** An integrated 0.26–1.7 kg unit with 0.2 A m^2 torquers, a 5–15 mN ms wheel set and a star tracker meets every flight need computed here. The in-house rods, wheel and printed-field star camera remain useful on the testbed only.
2. **Size torquers to the orbit, not to the test coil.** About 0.2 A m^2 per axis covers the high disturbance case with margin and detumbles from 10°/s within about two orbits. R-MTQ-1 ($\geq$1.0 A m^2) is a testbed requirement, set by the 90 μT coil and the bearing friction, and should not be carried to flight.
3. **Treat momentum management as mandatory.** A small wheel saturates within one to three orbits in the secular worst case, so a closed-loop magnetorquer dump (Section 8) is required from the start, and the residual dipole should be measured and budgeted.
4. **Duty-cycle the payload.** With body-mounted cells, imaging closes only at about one tenth of the orbit; either the payload computer is changed or deployable panels are added.
5. **Decimate ADCS telemetry and plan an S-band link if images matter.** A UHF link returns the decimated ADCS log, but only one image every few days.
6. **Upgrade the testbed toward a FlatSat.** A balanced three-axis bearing ($\leq 5 \times 10^{-5}$ N m as threshold, 10^{-5} as goal), a cage uniform to 5% at the orbit field, and an orbit simulation driving the sensors would let the test campaign verify B-dot and momentum management, which the one-axis platform cannot.
7. **Target an ISS-class orbit first.** At 400 km the ground passes are about 1.6× longer per day than at

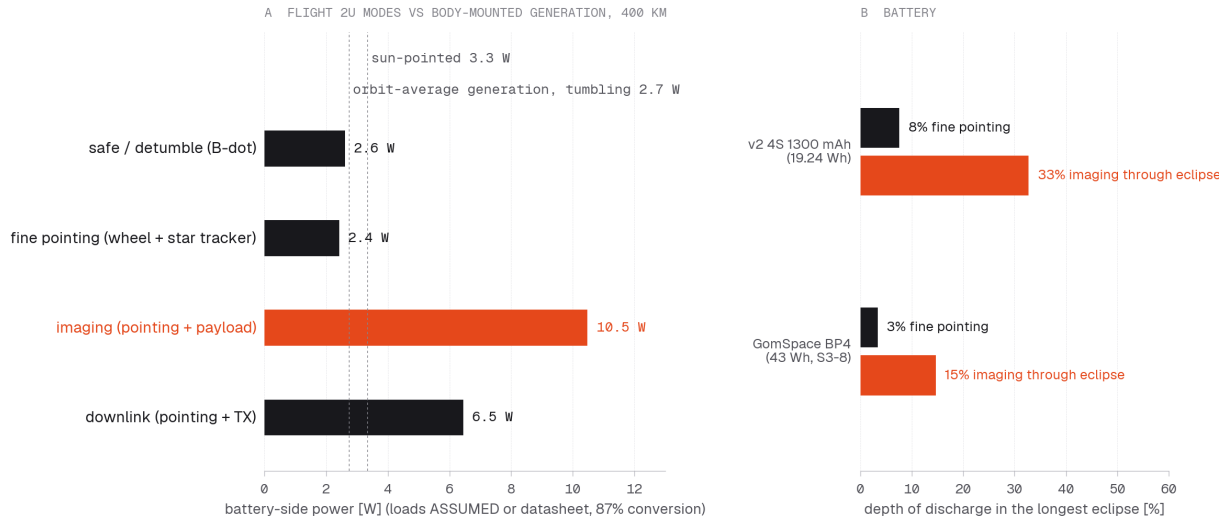


Figure 21: (a) Battery-side power per mode for the flight 2U against the orbit-average generation of body-mounted cells at 400 km; (b) depth of discharge in the longest eclipse for two battery packs. Imaging must be duty-cycled to about 10% of the orbit. Loads from datasheets [20] or ASSUMED; cell and loss factors ASSUMED.

500 km SSO for a mid-latitude station, and the higher drag shortens orbital life, both of which suit a first student flight.

15 Next-Version Hardware and Firmware: Design and Simulation

The goal of this section is to design the next version of the testbed so that the spacecraft stops on the commanded angle, and to quantify how well it could do so. The design has not been built. Every result below comes from simulation, and no part has been purchased, assembled or measured. The analysis links the following:

1. the identified bearing and wheel models of Section 4, extended with stiction, cogging and sensor limits near zero wheel speed;
2. a selection of real components with the source of every value recorded (`research/parts_v3hw.json`);
3. a stack of printed circuit boards (PCBs) used as structural levels, with a bus, a wire list and CAD including the wiring;
4. firmware written in C, run unchanged against the plant in a software-in-the-loop (SIL) harness.

15.1 Why the As-Built Wheel Could Not Stop on a Point

Three properties of the as-built drive combine. The first is resolution, which the report identified: one 13.67 RPM command step moves the body by $I_w/I_t \times 13.67$ RPM $= 0.058$ rad/s, which is nine times the 6.2 mrad/s imaging limit of Section 4. The second is authority, which the report did not identify. The acceleration-limited drive of Section 4 gives 0.58 mN m, only 1.4 times the Coulomb torque of the bearing and 0.8 times its upper 95% bound. A wheel with that margin cannot reliably break a stuck bearing free, and it has almost no torque left to shape a stop.

The third is the behaviour near zero speed, and this is where the question of direction enters. The electronic speed controller (ESC) label is only partly legible, so its model and firmware are unconfirmed, but sensorless commutation needs back-EMF in any firmware, and BLHeli_32 documentation states that it limits low-speed power so that the back-EMF can be detected [53]. The start-up kick in Fig. 38 ($A = 3.27$ rad/s-equivalent in 0.42 s) corresponds to a rotor speed of about 1550 RPM, which we take as the minimum controllable speed. Below it the drive has no authority, and the MCU has no wheel-speed feedback at all. Now consider a rest-to-rest slew. The bearing absorbs momentum throughout, so the wheel ends the manoeuvre with $\Omega_{end} = T_c\, t_{slew}/I_w$, about 314 RPM for a 6 s slew. A wheel started from rest therefore arrives at the end of the slew inside its own dead band, at exactly the moment fine torque is needed.

It is worth stating what this does not imply. A one-way wheel can torque the body in both directions by

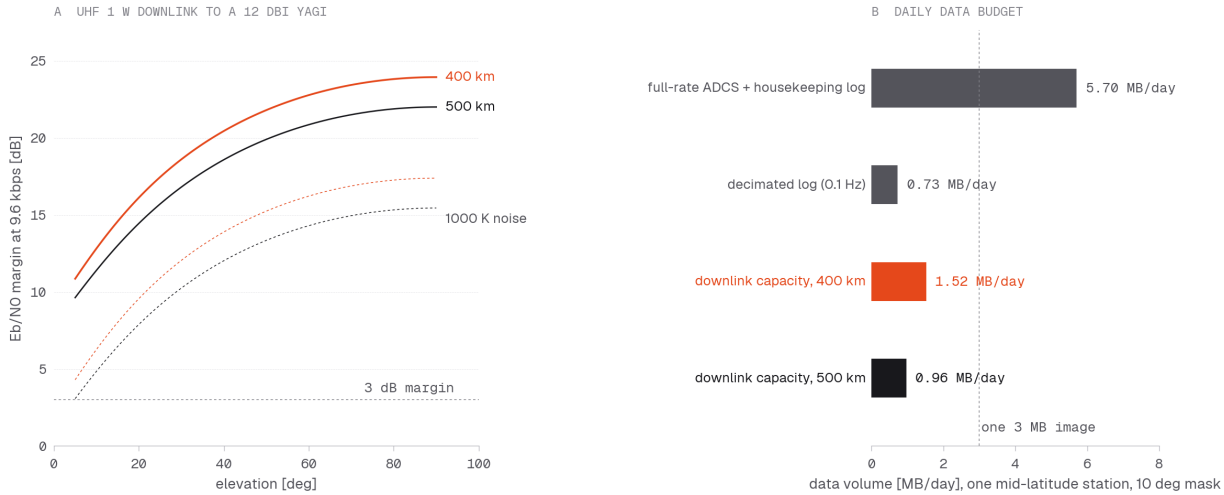


Figure 22: (a) UHF downlink margin at 9.6 kbps versus elevation for the two orbits (dashed: 1000 K system noise); (b) daily data produced and returned for one ground station. The link closes with margin, but full-rate telemetry exceeds the daily capacity 3.8×. Link equation and noise temperature from SMAD pp. 554–559; radio from [45]; passes simulated.

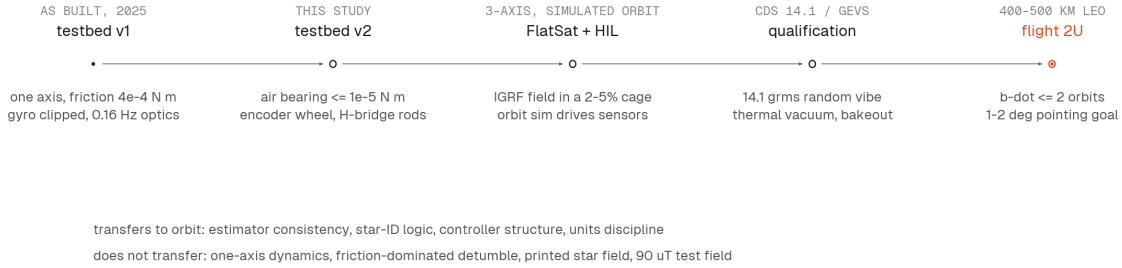


Figure 23: Path from the one-axis testbed to a flight 2U, with what each step verifies. Only the estimator, star-ID logic and controller structure transfer directly from the testbed to orbit. Test levels from CDS Rev. 14.1 [38] and GEVS [39].

running around a speed bias, so bidirectional rotation alone was not the missing feature. What was missing was torque authority, fine resolution and a reference that survives near zero speed. A bidirectional sensorless ESC adds a reversal but keeps the hole, and no document we read states the width of its dead band.

15.2 Drive Selection and the Zero-Crossing Model

We command the wheel through its rotor angle. The firmware integrates a speed reference Ω_r through zero, and the driver integrates Ω_r into an angle reference ϕ_r, so that

$$T_{em} = T_{ff} + g\,k_p(\phi_r - \hat{\phi}) + g\,k_d(\Omega_r - \hat{\Omega}) + \int g^2 k_i(\phi_r - \hat{\phi})\,dt, \tag{10}$$

where T_{ff} is the feed-forward, $\hat{\phi}$ and $\hat{\Omega}$ come from an on-axis magnetic encoder through a tracking filter, and g is a gain schedule. There is no speed at which the reference is lost. Stiction only delays the rotor until the position error has built enough torque, and cogging is fought by the same loop. The gain is 1 below 190 RPM and 0.15 above 380 RPM, where encoder noise matters more than stiffness. The g^2 on the integrator is required. With k_i left constant the third-order rotor loop violates $k_p k_d > I_w k_i$ at low gain, and the SIL showed a limit cycle of $\pm 0.35°$ at the body before this was found.

The truth model separates the rotor and the body:

$$I_w(\dot{\omega} + \dot{\Omega}) = \tau, \qquad J\dot{\omega} = -\tau + T_b(\omega) + T_m, \qquad \tau = T_{em} + T_{cog}(\phi) - T_w(\Omega), \tag{11}$$

where τ acts between rotor and stator, $T_{cog} = A_c \sin(84\,\phi + \psi)$ for the 12-slot, 14-pole motor, T_m is the torquer torque, and T_w and T_b are friction laws with Coulomb, Stribeck and viscous parts. Both interfaces use a stick test with exact capture at zero speed, so that the rotor can lock to the stator and the body can lock to the bearing. The plant runs at 10 kHz with encoder noise and quantisation, a current-loop lag and current noise, under firmware at 1 kHz. The bearing Coulomb torque is drawn from the bootstrap interval of Section 4; the break-away ratio (1.1–1.8), Stribeck speed, cogging amplitude (0.3–3 mN m), wheel friction and current noise are assumed ranges. No motor vendor we read publishes a cogging torque, so this range is an estimate and test T4 in the firmware notes replaces it with a measurement. Wheel friction is centred on the 0.25 mN m static friction published for a comparable 22 mm motor [54].

To isolate the crossing, the body tracks a slow 20 mrad/s ramp. The wheel must then supply the bearing torque steadily, so its speed runs through zero at a known time, and a perfect drive leaves the ramp undisturbed. Figure 24 shows the result and Table 14 summarizes it. In torque mode the rotor sticks at zero speed for about 0.7 s, the torque path is lost, and the body stops: the upset is 28 mrad/s, and an anti-cogging table does not help (20 mrad/s), because the fault is stiction and not cogging. Rotor-angle tracking reduces the upset to 5.8 mrad/s [5.6, 6.0]. Resolution beyond 12 bits changes little, since the sensor noise limits the loop before the quantisation does. Hall sensors alone fail (137 mrad/s, a clean crossing in only 50% of trials), and a sensorless bidirectional ESC crosses cleanly in 27% of trials. Flight data show why rare events matter: on the INTEGRAL spare wheel about 10% of crossings reached 30 mN m against a 5.4 mN m mean [51], and an air-bearing test by Carrara et al. found a ±38 mA dead zone that produced nearly 2° of error [52]. With wheel stiction at six times its Coulomb level we find 16 mrad/s (95th percentile 20), which is the worst case of the design.

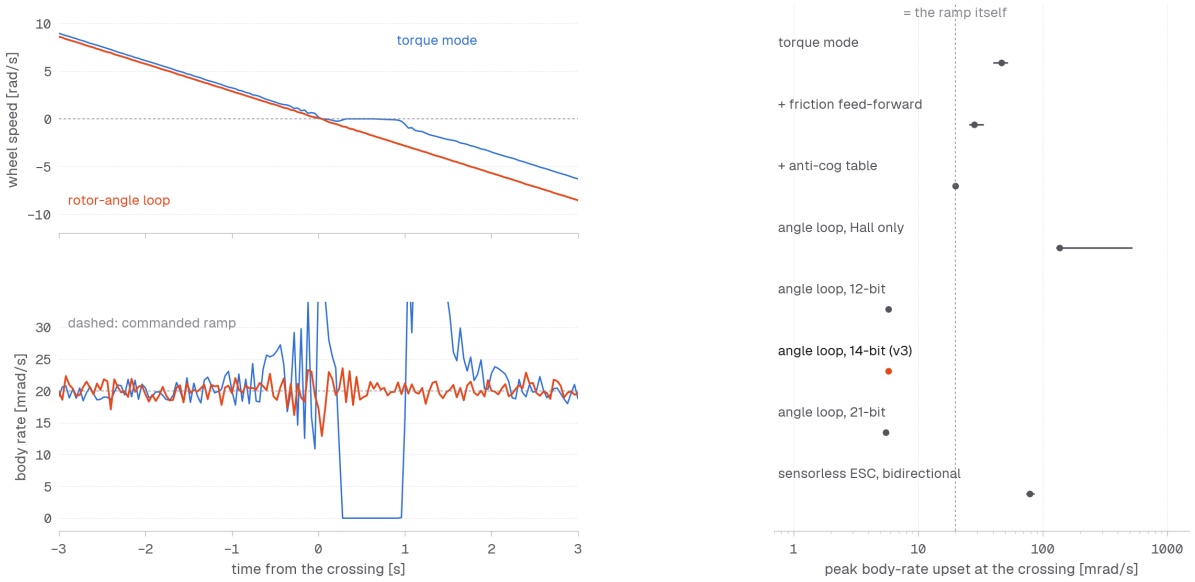


Figure 24: Wheel speed and body rate through a zero crossing on a 20 mrad/s ramp, and the peak upset for eight drive options. Rotor-angle tracking keeps the body moving where torque mode lets it stop. Simulation, N=400 per drive, 95% intervals.

15.3 Head-to-Head: Stopping on a Point

We now compare the designs on the task itself: a rest-to-rest slew of at least 20° inside the star field, for 60 s, $N = 1000$ per case. All cases run the same firmware core, and the as-built wheel is given a cascade tuned by grid search, so the comparison is against its best case. Figure 25 shows one matched example, which is a trial where the as-built wheel does break free; Figure 26 and Table 13 give the statistics.

The v3 design stops with a median error of 0.030° [0.028, 0.031] and a 95th percentile of 0.064°. It reaches 1° in 1.49 s [1.46, 1.53] after the command, and 0.1° in a median of 5.1 s. The as-built wheel with its best-case start has a median error of 21.6° [20.1, 22.8], and with the wheel off at the start, as in the report's test, 45°, because the arming kick spins the body. The as-built result is bimodal: in the 47% of trials where the break-away torque is below the wheel's limit the median error is 0.9°, and otherwise the body does not move

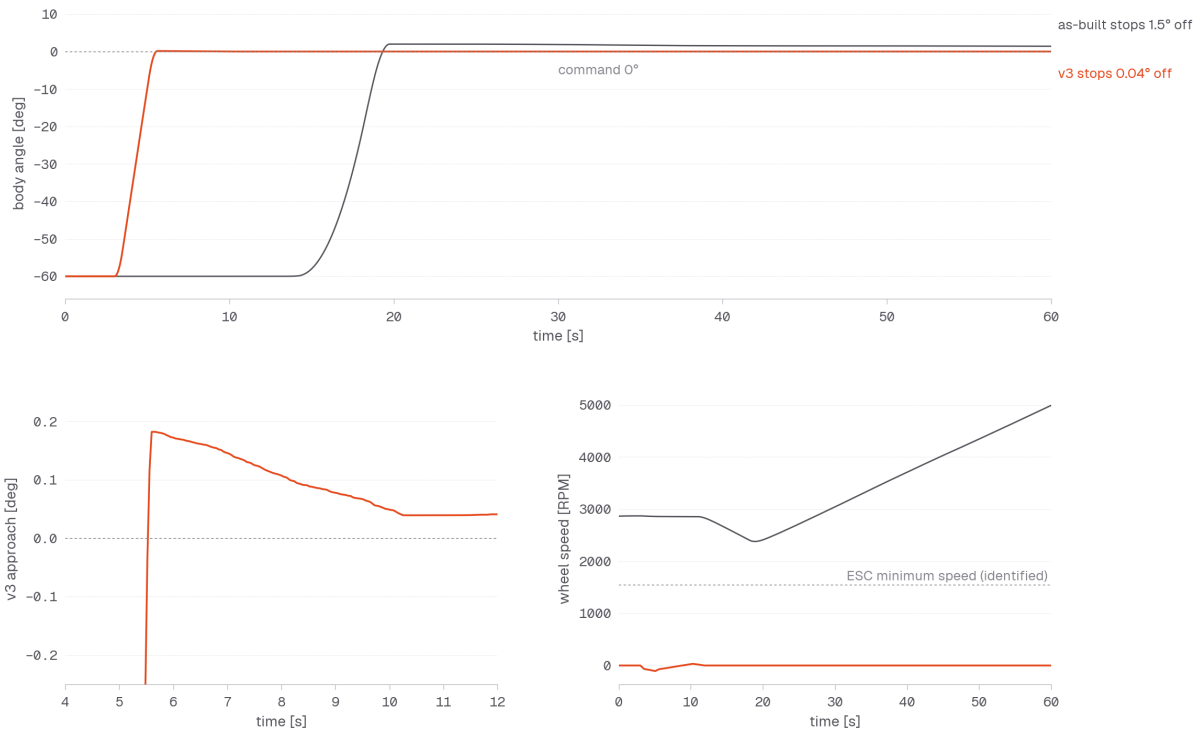


Figure 25: Body angle and wheel speed for the same 60° command on the same bearing. The as-built wheel accepts the command after 11 s, overshoots, and then sits still on the bearing while its speed winds up. Simulation; the v3 hardware does not exist.

at all (34°). A bidirectional ESC on the same motor is worse (84°), since each reversal passes through the dead band with a restart.

The factorial cases show which change matters. Better sensors on the as-built wheel leave the median at 21.5°, because the limit is torque. The v3 wheel on as-built sensors reaches 0.50° when it keeps lock, but only 66% of trials settle within 1° (95th percentile 147°), because fixes every 6.6 s with 4 s latency lose attitude after a fast slew. The wheel is the first-order change and the sensors are needed to keep it.

The stop-dead logic is a trade and not a pure gain, and this warrants discussion. The full design stops when the error is inside a band of 0.03° and the rate is below 1.5 mrad/s, removes the wheel torque, and sets the driver gains to zero, so that the bearing holds the body and encoder noise cannot shake it. It re-engages outside 0.08°. At break-away, the stiction step is removed from the integrator; zeroing the integrator instead made the body creep, and leaving it threw the body past the target. Without this logic the error is smaller (0.016°), but the body hunts for ever, with a limit cycle of 0.023° peak-to-peak (95th percentile 0.082°) and a residual rate of 0.25 mrad/s. With it, the rate is exactly zero over the last 5 s in 72% of trials [70, 75], and the limit cycle has a median of zero and a 95th percentile of 0.094°. Since the mission is imaging, we keep the stop.

15.4 What Accuracy Is Achievable on This Testbed

Figure 27 shows what sets the result. The final error follows the stop band, from 0.017° at a 0.02° band to 0.088° at 0.1°, so the headline number is largely a choice. White star-fix noise up to 1° hardly changes the median (0.048°), because a gyro at 0.6 mdps/$\sqrt{\text{Hz}}$ with 2 Hz fixes averages it away; the filter's estimate is good to 0.009° rms. The design fails in three identified ways: wheel torque below about 1.5 mN m (95th percentile 35° at 0.8 mN m), fixes slower than about 2 s, and latency above about 2 s.

This does not mean that 0.03° is achievable on the real testbed. The simulation omits the error sources that will dominate: star-map survey error, camera alignment, lens distortion, and the 1.5° sun-sensor residual found in Section 6. In our judgement, which is an estimate and not a result, 0.1–0.3° against a truth encoder is plausible once the star map has been calibrated, and about 1° before that. The stronger claim concerns rate. The body stops on the bearing, so its rate is far below the 6.2 mrad/s imaging limit, which was the

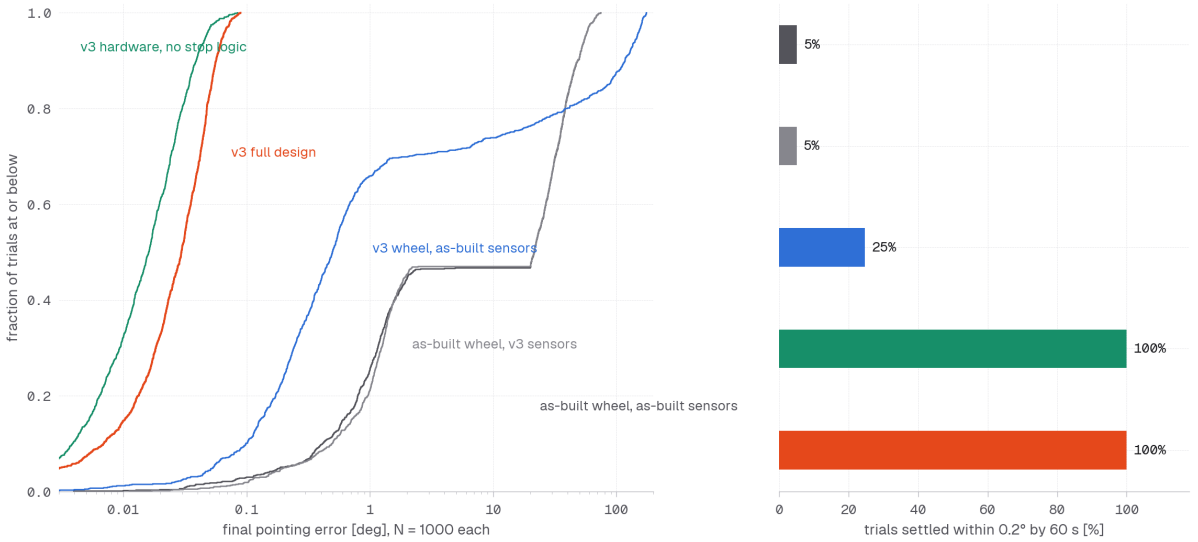


Figure 26: Final pointing error over 1000 simulated trials per case, and the share settled within 0.2°. The plateau of the as-built curves is the set of trials in which the wheel cannot break the bearing free. Simulation.

original objective. It is also notable that the friction which spoils the torque budget is what makes the dead stop easy. At 0.1 mN m of bearing friction the error is similar (0.052°), but only 0% of trials stop exactly, so a move to an air bearing would need active hold.

15.5 Mission Modes

The full firmware was also run from a spinning start. From 0.5–1.7 rev/s at an unknown angle, 100% of trials [lower bound 99%] reach 1° of the target, in a median of 9.7 s [8.7, 10.6], with a peak wheel speed of 925 RPM. Eight successive ±20° steps produce 7.9 commanded crossings per run, and all settle within 0.2° with a final error of 0.033°. In a momentum dump from 2670 to 570 RPM, the torquers alone take 339 s [323, 382], which agrees with $\Delta H/(mB)$. On this testbed the bearing does the work instead: slowing the wheel with a torque below stiction takes 122 s, and letting it coast in quiet hold takes 89 s, so the torquers cannot be shown to matter here, as the audit predicted. The firmware keeps that anchored path behind a configuration value that is zero for flight.

The estimator is consistent with the v3 sensors: mean NIS 0.998 (interval 0.988–1.012) and mean NEES 2.12 (interval 1.83–2.18), $N = 500$. With as-built sensors it is not (NIS 1.83, NEES 7.7), because 6% misidentified fixes are not Gaussian noise, so this is a statement about the sensors and not only the filter.

15.6 Component Selection

Table 15 compares wheel motors. The bus must supply 0.04 N ms (R-WHL-1) with the rim mass inside the budget, and the deciding constraint is speed at the bus voltage: gimbal motors of 60–160 RPM/V cannot reach the speed needed and require a motor and rotor of 250 g or more, whereas drone motors have the speed but are built for thrust. We select the CubeMars GL30 KV290 [55]: 41 g, 38 mN m/A, a 6 mm through bore for the encoder magnet, and $51. The maxon ECX FLAT 22 L [56] would save about 90 g through a smaller rotor at 7800 RPM, at the cost of imbalance force and a less convenient encoder path, and is a reasonable alternative. A cup-shaped brass rim of 70/52/10 mm around the motor gives $I_w = 1.46 \times 10^{-4}$ kg m^2, 0.046 N ms at 3000 RPM, and absorbs a 1.7 rev/s spin at 1310 RPM. The torque limit of 8 mN m is 19 times the bearing's Coulomb torque and 11.6 times its upper bound, against 1.4 and 0.8 for the as-built drive. The back-EMF at 3000 RPM is 11.8 V on a 14.8 V bus.

The driver is the mjbots moteus-c1 [57]: 8.9 g, 15–30 kHz loops, an integrated absolute encoder and CAN-FD at 5 Mbit/s. The ODrive Micro has a 12-bit encoder, which Table 14 shows would be adequate, but it was sold out when checked. Figure 28 shows the gyro trade. The Murata SCH16T-K01 [58] is about 5 times quieter than the ICM-42688-P [59] and specifies a bias instability of 0.5 deg/h, which no consumer part does, but its range is 300 dps. We therefore carry both: the wide gyro at ±2000 dps for detumble, the fine gyro

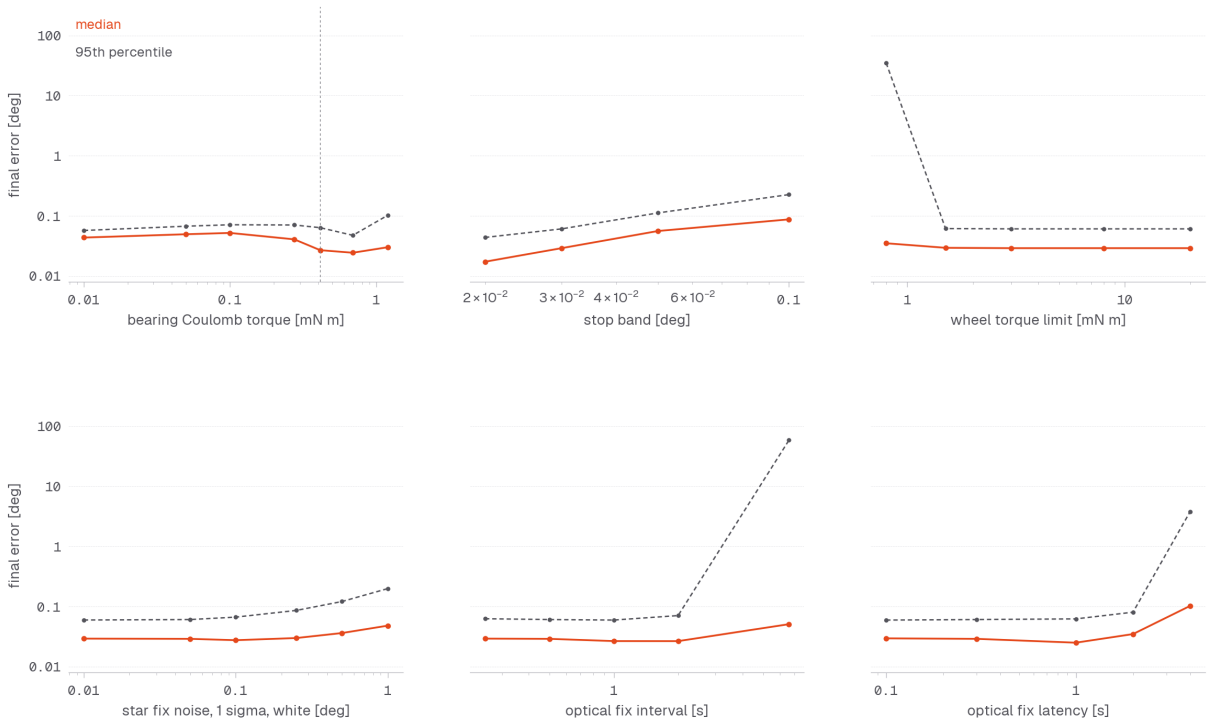


Figure 27: Final error versus bearing friction, stop band, wheel torque limit and optical noise, interval and latency. The stop band sets the error until a hard limit is crossed. Dotted line in the first panel: identified friction. Simulation, N=300 per point.

below 250 dps, and a running offset between them so that the hand-over does not step the estimate. Neither has a major-brand breakout, so both go on the avionics board; this also places them on the spin axis. The MMC5983MA [60] replaces the failed magnetometer parts at 40 nT rms. It sits on the top board, 118 mm from the torquers, where a 1.14 A m^2 rod still produces about 139 μT, so it is read in a torquer-off window.

A Teensy 4.1 [61] runs the real-time task, with hardware double precision, one CAN-FD port, three watchdogs and a microSD socket for logging. The Raspberry Pi 5 stays for vision only: it can run two cameras at once, which we could not verify for a Zero 2 W, and removing it from the control loop means a Linux stall can no longer disturb the attitude loop. The cameras are reused. The MLX90640 moves to its own 1 MHz bus so that the 6.6 s sampling of Section 6 can reach 2 Hz or better; the bus speed needed at the higher rates is unverified.

The relays are replaced by two TI DRV8876 bridges [62], which are bipolar, proportional at 20 kHz, and report coil current through IPROPI. On the reused 3.4 Ω rods at 14.8 V, 1.47 A needs a duty of 41% with about 29 mA of ripple. The coil heats at 0.67 K/s, so a 40 K limit allows 60 s of continuous drive, and each bridge dissipates 1.5 W at that current. The torque at 90 μT is 10.3×10^{-5} N m, or 0.25 of the bearing friction, which is the limit found in the audit and is unchanged by better electronics. Power comes from a 4S 1300 mAh pack, 80 g lighter than the 2200 mAh class, through a removable arm link and 10 A fuse, since Pololu warns that its push-button switch is not a safety disconnect [63]. An INA228 logs the pack, which the audit asked for after the unexplained 30 s run time.

Table 16 gives the bill of materials. The mass is 817 g, with a 95% interval of 752–883 g and 940 g with a 15% margin, against the 1 kg limit and 960 g as built. Only 43% of it is read from a source or computed, and the interval assumes ±30% on the remainder. The cost is about $827 against the $6,000 limit, excluding owned parts, and several prices are snippets. The battery power is 8.2 W in hold, 10.2 W in a slew and 23.4 W in a dump, with a peak of 1.8 A, so the pack gives about 2.9 h on the assumed duty, against the 30 s reached in the report's run. The stack is 157 mm tall against about 168 mm as built, so the spacecraft stays in the 1.5U class rather than the 2U proposed in Section 11.

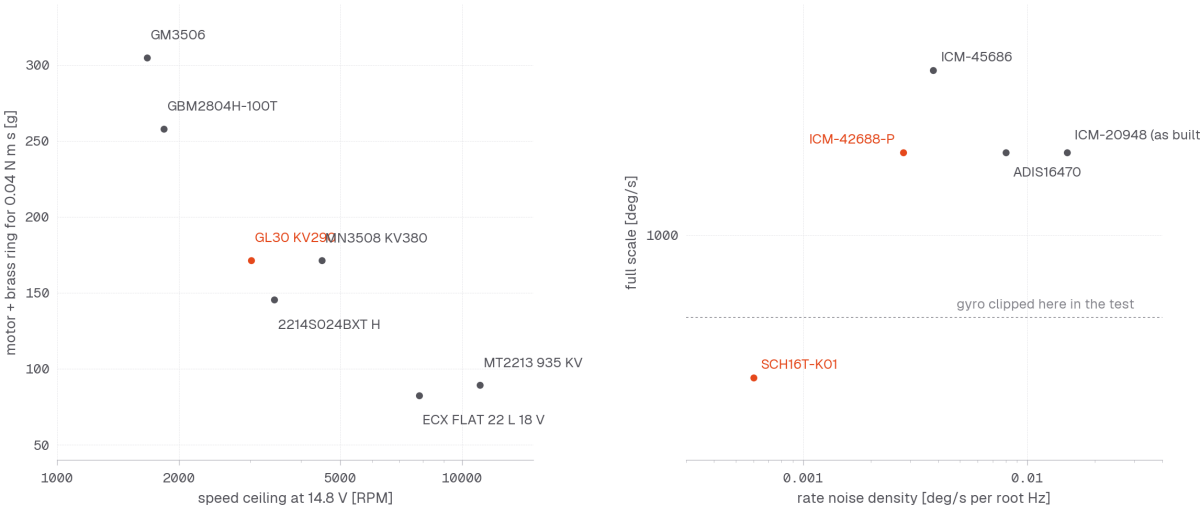


Figure 28: Wheel motors by the mass needed to store 0.04 N ms, and gyros by range and noise density. Selected parts are highlighted. Data sheet values, research/parts_v3hw.json.

Table 13: Stopping on a Commanded Point: Head-to-Head

Case	Final error, median [deg]	95th pct [deg]	Settled < 1°	Settled < 0.2°	Limit cycle p-p [deg]
As-built wheel and sensors, wheel off at start	45.2 [43.1, 47.9]	145	0%	0%	0.000
As-built wheel and sensors, wheel spinning	21.6 [20.1, 22.8]	55.8	25%	5%	0.000
Bidirectional ESC, as-built sensors	84.4 [81.6, 88]	162	0%	0%	0.000
As-built wheel, v3 sensors	21.5 [11.2, 22.7]	55.8	21%	5%	0.000
v3 wheel, as-built sensors	0.497 [0.447, 0.543]	147	66%	25%	0.000
v3 hardware, torque mode	0.0303 [0.029, 0.0315]	0.0758	100%	100%	0.000
v3 hardware, no stop-dead logic	0.016 [0.0149, 0.0171]	0.0465	100%	100%	0.023
v3 full design	0.0295 [0.0276, 0.0313]	0.0645	100%	100%	0.000

Simulation, N=1000 per case, 60 s, brackets are 95% bootstrap intervals of the median. Limit cycle is the median peak-to-peak true angle over the last 20 s.

Table 16: Bill of Materials (design, nothing purchased)

Lvl	Item	Part	Qty	Mass [g]	Cost [$]	Basis
L0	Base plate	PETG 100x100x4 (as built, reused)	1	50.8	5	D/A
L0	Wheel motor	CubeMars GL30 KV290	1	41.0	51	R/R
L0	Wheel driver	mjbots moteus-c1	1	8.9	175	R/S
L0	Wheel rim	Brass ring 70 OD, 52 ID, 10 thick, machined	1	146.6	60	D/A
L0	Wheel web and hub	Al web 2 mm, bolts to the GL30 rotor	1	14.4	30	D/A
L0	Motor pedestal	Printed PETG ring, carries motor over the driver	1	10.0	2	A/A
L0	Encoder magnet	Diametric magnet in the GL30 bore (moteus kit)	1	1.0	5	A/A
L0	Magnetorquer	Reused first-generation rods, 493 turns, 1.14 A m^2 at 1.47 A	2	44.8	owned	D/O
L1	PCB, power	100x100x1.6 4-layer blue FR4, 2 oz	1	31.0	20	A/S
L1	5 V regulator	Pololu D24V50F5	1	8.0	33	A/R
L1	Bus monitor	TI INA228 + 2 mOhm shunt	1	0.5	15	A/R
L1	Reverse/ideal diode	TI LM74700-Q1 + N-FET	1	0.5	4	A/A
L1	Torquer H-bridge	TI DRV8876 HTSSOP	2	0.4	16	A/R
L1	Arm plug and fuse	Removable XT30 arm link + 10 A blade fuse	1	6.0	6	A/A
L2	PCB, avionics	100x100x1.6 4-layer blue FR4	1	31.0	20	A/S
L2	Real-time computer	PJRC Teensy 4.1	1	11.0	32	A/A
L2	Fine gyro	Murata SCH16T-K01	1	1.5	83	A/S
L2	Wide gyro	TDK ICM-42688-P	1	0.1	12	A/A
L2	CAN-FD transceiver	TI TCAN1042 G variant	1	0.2	2	A/A
L2	Thermal sun sensor	Adafruit 4407 MLX90640 55 deg, reused	1	3.5	75	R/R

Lvl	Item	Part	Qty	Mass [g]	Cost [$]	Basis
L3	PCB, payload	100x100x1.6 2-layer blue FR4	1	29.0	5	A/S
L3	Linux computer	Raspberry Pi 5, reused	1	46.0	owned	A/O
L3	Star camera	Camera Module 3 Wide, reused	1	4.0	owned	A/O
L3	Science camera	HQ Camera M12 with lens, reused	1	45.0	owned	A/O
L3	Camera brackets	Printed PETG, slotted	3	9.0	3	A/A
L4	PCB, top	100x100x1.6 2-layer blue FR4	1	29.0	5	A/S
L4	Battery	Tattu 1300 mAh 4S 75C	1	156.0	40	S/A
L4	Magnetometer	SparkFun SEN-19921 MMC5983MA	1	1.5	18	A/R
L4	Battery strap	Hook strap + printed cradle	1	8.0	3	A/A
Stack	Stack connector	Samtec ESQ-126-39-G-D	3	13.5	42	A/A
Stack	Corner standoffs	Al hex M3, four corners, total 465 mm (CAD)	1	24.0	16	D/A
Stack	Fasteners	M3 screws, M2.5 Pi spacers	1	12.0	8	A/A
Stack	Harness	See wire list	1	19.3	25	D/A
Stack	Balance ballast	Trim allowance	1	10.0	0	A/A
Test	Rate-table truth encoder	AS5600 on the shaft, off the spacecraft	1	0.0	15	A/A
	Total			817	827	

Basis is mass/cost: R read from a data sheet or vendor page, S search snippet only, D derived, A assumed, O owned. The test-table encoder is off the spacecraft and carries no flight mass.

Table 17: Wire List

ID	From	To	Net	AWG	Length [mm]	I_{max} [A]	Drop [mV]
W01	L4 battery XT30	L1 J1 arm link	VBAT	16	112	6.00	8.9
W02	L4 battery XT30	L1 J1	GND_PWR	16	112	6.00	8.9
W03	L1 J2	L0 moteus J1	VBAT fused 5 A	22	65	1.50	5.1
W04	L1 J2	L0 moteus J1	GND_PWR	22	63	1.50	5.0
W05	L1 J3	L0 moteus J2	CAN-FD H	26	60	0.05	0.4
W06	L1 J3	L0 moteus J2	CAN-FD L	26	59	0.05	0.4
W07	L0 moteus	Motor phase A	phase	24	21	1.00	1.8
W08	L0 moteus	Motor phase B	phase	24	21	1.00	1.8
W09	L0 moteus	Motor phase C	phase	24	21	1.00	1.8
W10	L1 J4	Torquer X	bridge X out 1	22	50	1.47	3.9
W11	L1 J4	Torquer X	bridge X out 2	22	50	1.47	3.9
W12	L1 J5	Torquer Y	bridge Y out 1	22	48	1.47	3.8
W13	L1 J5	Torquer Y	bridge Y out 2	22	49	1.47	3.8
W14	L3 J6	L4 magnetometer	+3V3 and I2C (4 core)	28	68	0.02	0.3
W15	L3 J7	Pi 5 GPIO 2,4,6	+5V, GND	20	22	3.00	2.2
W16	L3 J8	Pi 5 GPIO 8,10	UART 921600	28	23	0.01	0.0
W17	L2 J9	MLX90640	I2C 1 MHz, 3V3 (4 core)	28	61	0.03	0.4
FFC1	Pi 5 CAM0	Star camera	22-pin FFC	FFC	106	0.00	0.0
FFC2	Pi 5 CAM1	Science camera	22-pin FFC	FFC	73	0.00	0.0

Lengths are the routed CAD path times 1.3 for service slack (assumed).

15.7 Structure and Wiring

As built. We re-read the photograph and the test clips before designing the stack, and they corrected the earlier CAD reconstruction. The one blue board visible is the power distribution board of Section 2.4.3 of the report. It is about 103 mm wide, is a structural level with the corner members through it, and sits just below plate 1, which the earlier model had wrong. The corner members are stacked brass standoffs rather than one rod, the wheel is about 66 mm across rather than 56, and the cameras can now be identified from their labels as a Camera Module 3 and an MLX90640. The other PCB level is the dark battery board; the remaining levels are aluminium. Figure 29 shows the corrected model with its wiring, all of it estimated.

Improved stack. The v3 design makes every level a PCB, so that power and data pass through the structure and the wire list shrinks to what must leave a board. There are four blue boards on M3 aluminium standoffs at the same 90 mm pitch, which at 2-layer and 4-layer weights are lighter than the 43 g aluminium plates they replace. The base stays printed, since it carries the table interface. Level L1 carries battery entry, arm

Table 14: Zero-Crossing Test: Body-Rate Upset on a 20 mrad/s Ramp

Drive	Peak upset, median [mrad/s]	95th pct	Crossed cleanly
Torque mode, no compensation	46.5 [40.2, 52.5]	602.1	100%
Torque mode, friction feed-forward	28.2 [25.6, 33.5]	63.6	100%
Torque mode, plus anti-cog table	20.0 [20.0, 20.0]	20.1	100%
Rotor-angle loop, Hall only (42 edges/rev)	136.6 [127.3, 521.1]	712.3	50%
Rotor-angle loop, 12-bit encoder	5.8 [5.6, 5.9]	8.7	100%
Rotor-angle loop, 14-bit encoder (v3)	5.8 [5.6, 6.0]	9.0	100%
Rotor-angle loop, 21-bit encoder	5.5 [5.3, 5.7]	8.3	100%
Sensorless ESC, bidirectional	78.4 [73.7, 85.4]	3293.8	27%

Simulation, N=400 per drive. An upset of 20 mrad/s means the body stopped. 'Crossed' means the wheel passed through zero once in the window; failures stalled or chattered.

Table 15: Wheel Motor Trade at a 14.8 V Bus

Motor	KV [RPM/V]	Ceiling [RPM]	Motor + ring [g]	Price [$]	Note
CubeMars GL30 KV290 (table 255)	255	3019	172	50.99	6 mm through bore, 'low cogging', no number
maxon ECX FLAT 22 L 18 V	661	7826	83	203.0	EUR 188.20, no through bore read; Hall version
Faulhaber 2214S024BXT H	290	3438	145	–	C0 0.25 mN m; price not shown
CubeMars GL40 KV70 (table 63.5)	64	752	634	72.99	14 pole pairs, far too slow at 14.8 V
iPower GBM2804H-100T	155	1835	258	18.99	KV from 1468-1622 RPM at 10 V
iPower GM3506	141	1669	305	23.99	KV from 2149-2375 RPM at 16 V
T-Motor MN3508 KV380	380	4499	171	69.9	Kt from KV; drone motor
Emax MT2213 935 KV (as-built class)	935	11070	89	–	iron-core drone motor

Ceiling is 80% of no-load speed. Mass is the motor plus a brass ring at 61 mm mean diameter storing 0.04 N m s. No vendor read publishes a cogging torque.

link, 5 V regulator, monitor and bridges. L2 carries the Teensy and both gyros on the spin axis. L3 carries the Pi and cameras, and L4 the pack and magnetometer. An ESQ-126 2×26 stacking connector [64] joins L1, L2 and L3 at 15.24 mm spacing; its 5.7 A pin rating is for one powered pin per row, so we derate it and use eight pins each for battery and return. Table 18 gives the pinout, and Figure 30 and Table 17 give the wiring. The largest drop on any wire is 9 mV.

Grounding is a star at L1. The motor and bridge returns run on GND_PWR, and the logic, gyros and CAN reference run on GND_SIG, joined once at the shunt so that wheel and torquer currents do not flow through the gyro reference. Current paths are short on purpose: the wheel's current enters at L1 and leaves to the base through a dedicated pair, and each torquer's out and return run as a twisted pair so that the loop area, and with it the field at the magnetometer, stays small. Wires leaving a board are tied at the standoffs within 20 mm, so that a pull cannot reach a solder joint or crimp, and each level unplugs without cutting anything.

The wheel is on the geometric axis. Its rim sits above the torquers and below L1, with a 37 mm keep-out checked against every wire. A balance grade of G6.3 at 3000 RPM permits 20 μm of offset and 0.31 N of rotating force, and G2.5 permits 8 μm and 0.12 N. Twelve blind holes in the lower rim face take the trim. The wheel's axis must also pass through the table's axis; the design adds 10 g of ballast allowance for that. Figure 31 shows the stack exploded and the harness alone.

CAD. Both models are parametric CadQuery assemblies with estimated dimensions flagged part by part, the as-built with 79 parts and v3 with 134, with each wire a swept solid. A boolean check of every pair of parts found 0 interferences above 0.5 mm^3, including the wheel's swept volume. It also forced real changes: the second torquer moved 5 mm inward to clear the standoffs, and the motor pedestal needs windows for the driver's corners, leaving pillars of 2 mm whose stiffness has not been assessed. The CAD gives a non-rotating inertia of 9.43×10^{-4} kg m^2 against 7.93×10^{-4} from the lumped estimate. We use the CAD value, so the body inertia with the platform is 13.97×10^{-4} kg m^2, and the simulations span ±15%.

15.8 Firmware

The firmware is in `firmware_v3/`. Its portable core has no hardware calls: one function, `adcs_step()`, runs at 1 kHz and contains the estimator, the cascade, the wheel reference, the momentum logic and the state machine of Figure 32. The attitude loop runs at 200 Hz inside it. The controller is a profile generator, a proportional angle loop and a proportional-integral rate loop, with inertia feed-forward and a smoothed Coulomb term. The estimator is the [θ, b] filter of Section 6, ported to C, with one change: a filter now needs two agreeing fixes to initialise, so that a misidentified star cannot seed it. A single enum defines the

Table 18: Stack Bus Pinout (ESQ-126, 2×26)

Net	Pins
VBAT	A1, A2, A3, A4, A5, A6, A7, A8
GND_PWR	A9, A10, A11, A12, A13, A14, A15, A16
+5V	A17, A18, A19, A20
GND_SIG	A21, A22, A23, A24, A25, A26, B23, B24, B25, B26
+3V3	B1, B2
CANH	B3
CANL	B4
CAN_GND	B5, B6
UART_TX	B7
UART_RX	B8
I2C_SDA	B9
I2C_SCL	B10
PWM_X_EN	B11
PWM_X_PH	B12
PWM_Y_EN	B13
PWM_Y_PH	B14
IPROPI_X	B15
IPROPI_Y	B16
nFAULT	B17
nSLEEP	B18
ARM_SENSE	B19
WDT_FLAG	B20
SPARE	B21, B22

states for firmware, ground and logs, which closes audit item A27. Telemetry is a 49-byte frame with CRC16 and COBS framing at 50 Hz over a 921.6 kbaud UART, which uses 2.7% of the link, while the 115.2 kbaud as-built link carried 86-byte packets. The full 1 kHz record goes to microSD, and a 1 Hz decimation uses 5.1% of the 9.6 kbit/s flight link of Section 13.

Faults latch until cleared. Injected at 25 s in 40 runs each, loss of both gyros reached SAFE in 50 ms and loss of the wheel link in 100 ms, and in SAFE the wheel winds down below stiction so that the body stays put. A hung control task was reset by the watchdog in 40 of 40 runs, with the wheel driver's own command timeout as the second line of defence, and the spacecraft recovered to within 0.016°. Corrupted star fixes were gated, with a final error of 0.028°.

The SIL proved its worth by finding faults in this code, all now fixed: a boot calibration that stored a wheel transient as a 70 mrad/s false bias, an acquisition that could stop on an unconfirmed fix outside the field of view, the creeping integrator reset, and the rotor-loop instability above. There are 29,912 unit checks and eight physics and regression tests, among them momentum conservation through a slew, an analytic spin-down, and agreement with the Python simulator of Section 5. The core and drivers compile for Cortex-M7 to 26 kB of object code, but they have not been linked for a target or run on one. The Teensy hardware layer is not written, the SCH16T frames are a stub, and the moteus protocol was checked only against the worked example in its documentation. One bench item comes first: the register page does not say that a velocity command with a NaN position integrates the control position, and the rotor-angle scheme depends on it.

16 Discussion

16.1 Lessons Learned

The audit and redesign point to a small number of practices that would have changed the original outcome:

1. **Close the units loop.** Two critical errors (inertia ×2, RPM ×10) were found by recomputing every number; one unit check would have changed the actuator conclusion.
2. **A requirement must be able to fail.** Requirement A passed with the ADCS off. Every dynamic test needs an ADCS-off control run and a bearing whose disturbance is $\ll$ the control torque.
3. **Do not judge estimator quality by eye.** 'Inside the sigma bounds' accepts both a filter 90× over-confident and one 6× under-confident; NEES/NIS with chi-square intervals and deliberately mis-tuned controls should be used instead.

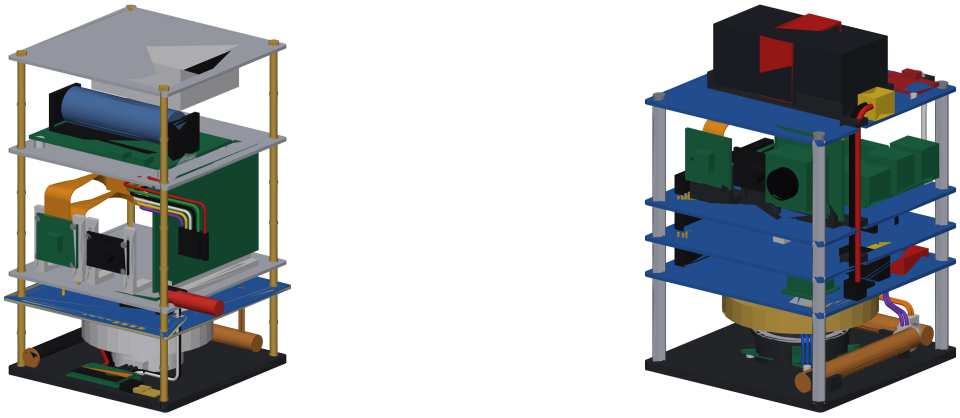


Figure 29: CAD of the as-built stack, corrected from the photograph (left, dimensions estimated) and of the v3 design (right, not built), both with routed wiring.

4. **Know the sensor limits before the test.** A saturated gyro, 6.4 s optical sampling and a coarse sun-sensor staircase were all visible in the data; range, rate and quantisation should be measured first.
5. **Log what will need to be proven.** Exposure, mode, wheel speed, voltage and the gyro full-scale flag were missing, and each cost an inference.
6. **Design for the physics bound.** The de-saturation requirement needed ~ 9 A m^2; momentum dumping is bounded by $H/(mB)$, and no controller changes that.
7. **Screen novelty honestly.** Nearly every candidate already existed; the value is in the quantified, reproducible application and in the negative results.
8. **Treat third-party code as data.** The repository's lookup table did not match its documented map; reading it, rather than running it, was enough to show this.

16.2 Sources of Error and Model Limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

(i) Digitised data. The Fig. 37/38 values are digitised (resolution ≈0.05 rad/s, 0.03 s), and the Hall trace is sparse. The identified friction and wheel parameters carry this resolution in their uncertainty bands.

(ii) Phenomenological wheel model. The wheel/throttle model is phenomenological and identifies only the acceleration limit; I_w/I_t separately is not identifiable.

(iii) Assumed parameters. The component masses, the powers (Pi 5 ≈7 W), the rod μ_r and the gyro bias random walk are assumed.

(iv) One-axis, idealised simulation. The simulations are one-axis and idealised: they include no cross-axis coupling, no rolling-shutter skew (readout time unverified), no COM unbalance on the bearing, and no Stribeck friction.

(v) Estimator consistency. EKF consistency is shown only for simulated Gaussian noise that matches the filter's model. The delay is handled by a conservative approximation, and real sensor noise (non-Gaussian, temperature-dependent) is not tested.

(vi) Simulated star images. The star-ID campaign uses simulated images and a catalogue-error model; the real lens distortion of the 102° lens is not modelled.

(vii) Rod sizing source. The Amin–Lightsey sizing equations [6] could not be read (access denied), so the rod model is first-principles only.

(viii) Costs. The costs of air bearings and strobes are not priced (unverified). The commercial ADCS prices are vendor snippets and launch prices change yearly, so the cost paragraph of Section 13 is an order of magnitude only.

(ix) Atmospheric density. No cited density table was retrieved (the shared search budget ran out), so the drag torque uses an exponential through SMAD's two quoted densities. This over-estimates drag at 400–500 km; since the aerodynamic term is already below the magnetic one and every term is at least 147× below the friction, no conclusion depends on it.

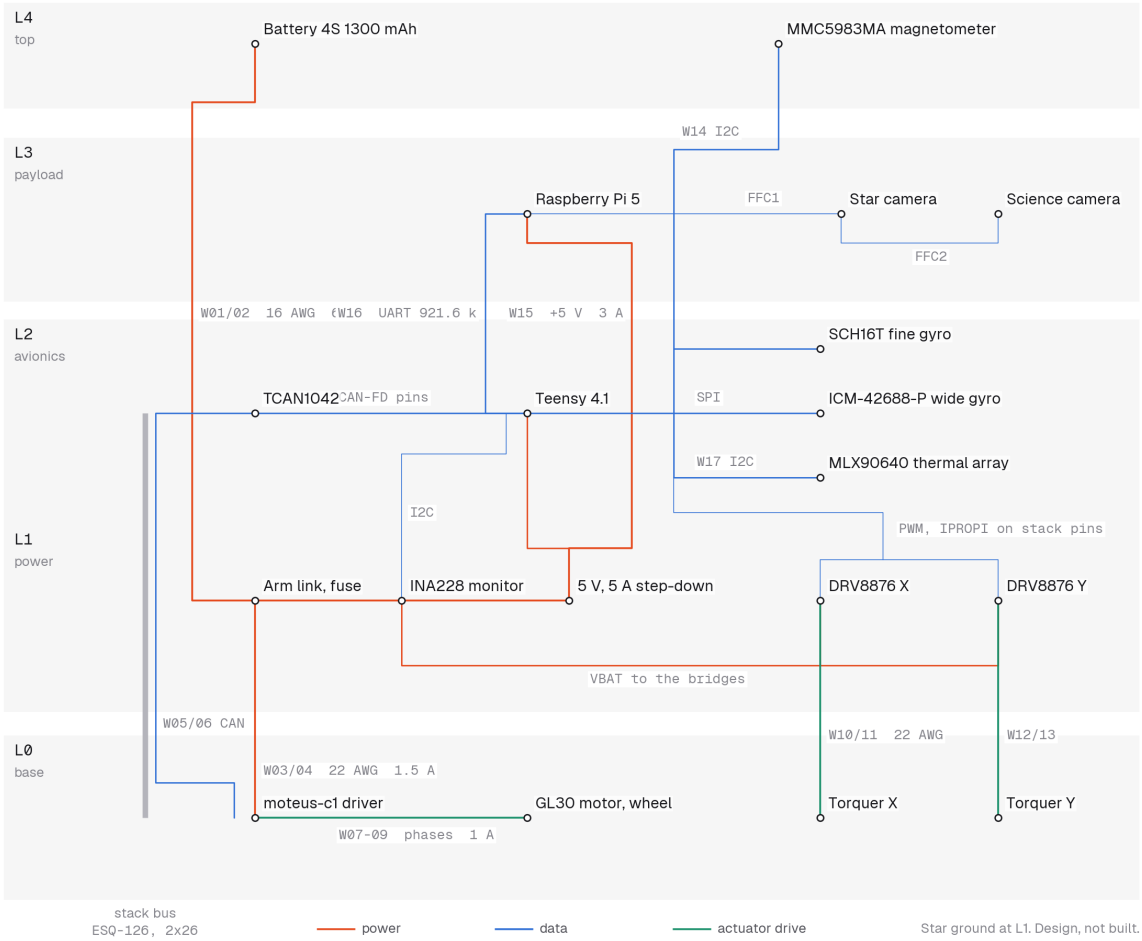


Figure 30: Block and wiring diagram of the v3 stack, one lane per PCB level, with wire numbers from Table 17. Design, not built.

(x) Field and orbit model. The geomagnetic field is an axial dipole (6.7% above IGRF-14 at the check point), the orbit is circular without J2 or drag, and the B-dot gain is one ASSUMED tuning. Detumble times are therefore indicative to perhaps a factor of two; the conclusions that the 0.2 A m^2 class suffices and that a larger dipole does not shorten detumble rest on the field geometry and are robust to these choices.

(xi) Flight loads and link parameters. The flight power budget mixes datasheet loads with ASSUMED computer, radio and payload powers, and the link uses ASSUMED antenna gains, losses and a single ground station. The imaging duty cycle and the image return rate move in proportion to these inputs; the conclusion that imaging, not the ADCS, drives power and data holds unless the payload power falls by a factor of several.

(xiii) v3 is a design. Nothing in Section 15 has been built or measured. Cogging, wheel stiction, current noise and the bearing break-away ratio are assumed ranges, and they drive the zero-crossing result; the ranking of the drives is likely to hold, while the absolute upsets are not.

(xiv) Omitted motion. The v3 simulation is one-axis, with no wobble, no centre-of-mass offset and no wheel imbalance. A 0.3 N rotating force at 50 Hz on a loaded bearing could change the friction law itself. This is the largest open question for the wheel.

(xv) Optical errors. Star fixes are unbiased white noise with an assumed 0.05° and 0.3 s latency. Real systematic errors would set the floor, so the final pointing numbers are upper bounds on performance and not predictions.

(xvi) Driver model. The driver is modelled from its documented structure, not its code. The behaviour under a NaN position, the absence of a k_i scale, and the encoder reading a magnet in this motor's bore are unverified, and any one of them could force a change to the rotor-angle scheme.

(xii) Benchmark bases. Published pointing figures use different statistics (1σ, 3σ, rms) and some are vendor claims, so the comparison of Section 12 orders classes and should not be read as a product ranking.

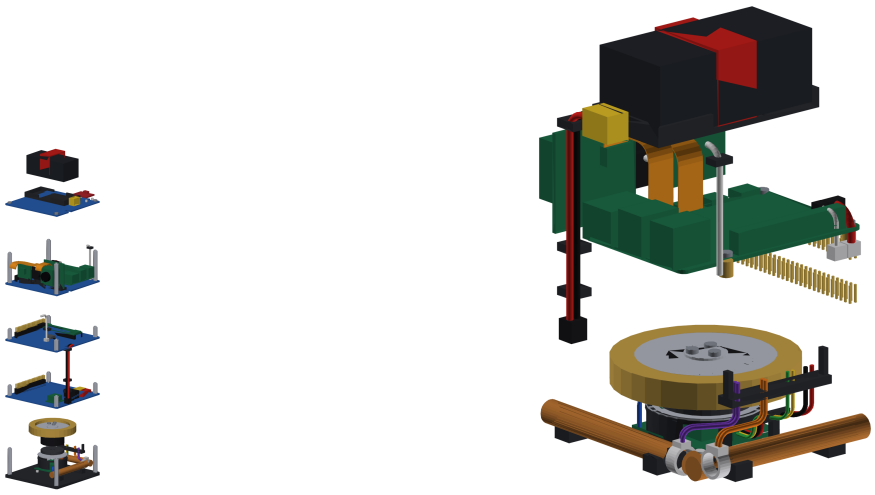


Figure 31: The v3 stack exploded along the spin axis (left), and the harness with the boards hidden (right). Design, not built.

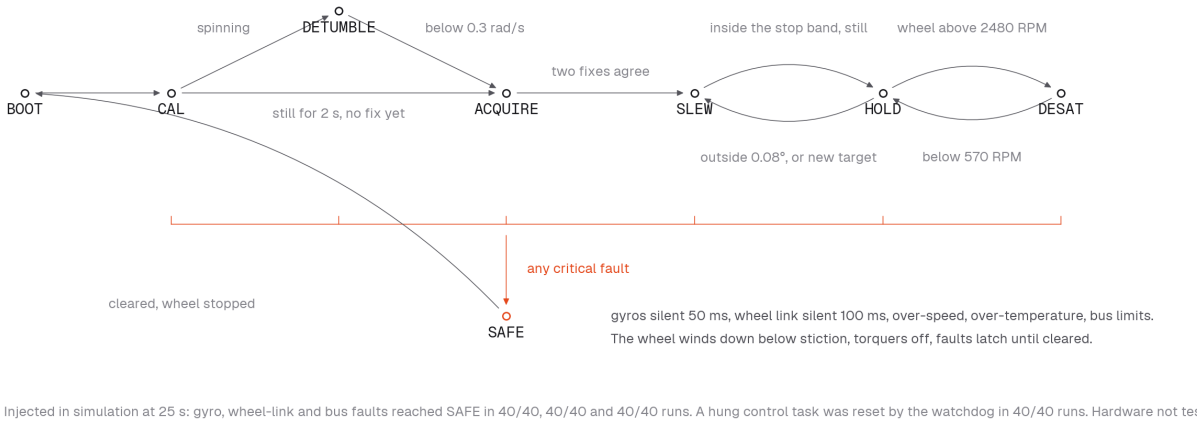


Figure 32: Firmware state machine with its fault paths. Faults were injected in simulation only.

17 Conclusion

This work re-examined the SPACE 584 Fall 2025 one-axis ADCS testbed by auditing the final report, identifying a dynamic model from the report's own digitised data, and redesigning the controller, estimator, star identification, magnetic subsystem and imaging strategy, each evaluated by deterministic Monte Carlo.

At the component level, the results show that several of the report's conclusions follow from arithmetic and unit errors rather than from the hardware: the inertia and wheel-resolution requirement were mis-sized, the gyro was saturated in the headline test, and the report's filter is not consistent even under its own assumptions. The identified model shows that the testbed's bearing friction dominates both the detumble requirement and the available magnetic torque, and that the wheel channel was acceleration-limited.

At the system level, a closed-loop rate/attitude controller with a consistent, delay-aware estimator raises the simulated probability of reaching imaging within 60 s from that of the report's state machine to near certainty on v2 hardware, while a time-optimal controller shortens rest-to-rest slews. Most candidate improvements are replications of published methods, and two do not hold.

Compared with current practice, the testbed's accuracy, bearing quality and sun sensor are well behind flight hardware and university facilities, so its value lies in verifying algorithms and test discipline rather than hardware. Sizing a flight version with the SMAD methods shows that a real orbit is far gentler in torque than the testbed but demands momentum management, that commercial-class torquers detumble the

spacecraft from realistic tip-off rates within a few orbits, and that the imaging payload, not the ADCS, sets the power and data budgets.

A next-version design was then developed and tested in software-in-the-loop. Torque authority and rotor-angle tracking through zero speed are what let the spacecraft stop, and the simulation shows the improvement of Section 15, within the 1 kg and $6,000 limits.

In general, the results demonstrate that the testbed, as built, could not have shown the behaviour its requirements asked for, and that the limiting factors were requirement design, units and sensor limits rather than control theory. Future work should follow the recommendations of Section 14: buy a flight-proven ADCS for any flight version and size it to the orbit, and build and run the v2 test campaign toward a three-axis FlatSat, starting with an air-bearing platform and an ADCS-off control run for every dynamic test, and should replace the assumed loads, rod permeability and sensor noise with measurements.

Acknowledgments

The author thanks the members of the SPACE 584 Fall 2025 class, whose final report, data and code repository are the basis of this re-analysis, and the course instruction for the original problem statement.

A Reproducibility

`run_all.py` regenerates every number, figure and table from the PDF and the data (the v3 flight sizing and comparison figures come from `analysis/a10_flight.py` and `analysis/a11_sitefigs.py`, with sources in `research/sources_v3.md` and site figures in `figs_site/`), with seeds derived from one master seed through named streams. `tests/test_all.py` checks the closed forms, the key results, bit-for-bit determinism and the power of the NIS test. The quick mode runs the same code with a smaller N.

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